CHAPTER

PLANT SECONDARY METABOLITES WITH HEPATOPROTECTIVE EFFICACY

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CHAPTER OUTLINE

3.1 Introduction 71
3.2 Liver Diseases and Their Overview 72
3.3 Physiological Role of Liver 73
3.3.1 Carbohydrate Metabolism 73
3.3.2 Fat Metabolism 74
3.3.3 Protein Metabolism and Storage of Vitamins 74
3.3.4 Neutralization of Toxic Substance 74
3.4 Mode of Hepatotoxicity 74
3.4.1 Reactive Metabolites Formation 74
3.4.2 Lipid Peroxidation and Redox Cycling 75
3.4.3 Effect of Chemical Agents on Major Cellular Systems 75
3.4.4 Modification in Calcium Homeostasis 75
3.4.5 Drug-Induced Hepatotoxicity 76
3.5 Phytochemicals 79
3.5.1 Current Status of Therapeutic Plants 80
3.5.2 Hepatoprotective Properties of Phytochemicals and Their Mode of Action 80
3.6 Conclusion 91
Acknowledgment 94
References 94
Further Reading 104

3.1 INTRODUCTION

The liver is the principal metabolic organ of the body that actively contributes in essential functions to regulate nearly all metabolic processes and homeostasis of the body. It is present in the upper right side of the abdomen with an average weight of 1.5 kg in a healthy person having a body weight of 75 kg. It performs more than 500 major functions including breakdown of food constituents to critical blood components, preservation of minerals and vitamins, production of essential

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plasma proteins and minerals, which are used in hormonal balance, metabolic processes, and detoxification of toxic compounds of the body (Ghany and Hoofnagle, 2005). Moreover, it is a major site for synthesizing blood-clotting factors which avert blood from clotting during circulation (Saleem et al., 2010). It also regulates the normal glucose level in fasting condition and promotes the glycogen metabolism (Michael et al., 2000). A wide range of studies suggested that the free radicals (reactive oxygen species (ROS)/RNS) exert oxidative stress which is a major cause of hepatic abnormalities like degeneration, necrosis apoptosis, swelling, etc. While in some other cases hepatic sicknesses are mostly triggered by chemicals and drugs when taken in very high quantities (Sharma et al., 2017), other causes of hepatic damage may include lipid peroxidation and covalent binding leading to consequent tissue injury. Free radicals such as peroxyl, superoxide, hydroxyl, and alkoxy radicals, etc. lead to damage of lipid membrane, protein, and nucleic acid (Singh et al., 2008; Pal et al., 2014). During hepatic injury the efficacy of natural antioxidant system is altered. Free radicals are produced by both from external factors (X-rays, UV radiation, pollutants) as well as internal factors (normal metabolic process in mitochondria). Intracellular concentration of free radicals is governed by both the generation by endogenous or exogenous factors and their elimination by various endogenous antioxidant system (enzymatic and nonenzymatic) (Haque et al., 2014).

Even though there had been appreciable developments in conventional medical treatment in the past 20 years, drugs existing for the treatment of hepatic abnormalities were restricted in efficiency and could have provoked numerous adverse side effects. In response to these reasons that limit the use of conventional drugs, efforts were continuously being made to develop new sources of agents with hepatoprotective potential (Mamat et al., 2013). Therefore, the use of alternative medicines, predominantly herbal/plant-based treatments, has grown globally due to its therapeutic potential and less toxicity, to cure various diseases. The plant kingdom is a natural treasure of herbal medicinal agents. In recent years various therapeutic plants were evaluated to check their hepatoprotective potential in animal models. Traditional medicines and their formulations with favorable results against numerous pathological disorder have recently been considered as alternative therapies (Biswas et al., 2014). A number of natural compounds have been isolated and identified for their physicochemical and pharmacological activities.

3.2 LIVER DISEASES AND THEIR OVERVIEW

Liver sickness is a dangerous disease that causes huge morbidity and mortality worldwide. It is a chief organ for degradation of drugs and chemicals, hence it protects living beings from toxic foreign materials. Xenobiotics, microorganisms, metabolic diseases, autoimmune diseases, inherited hepatic diseases, and hepatic malignancies are major causative agenst of hepatic abnormalities (Daniel, 2009). According to a study, about 1.3 million of the population of the world die due to viral hepatitis annually, over 350 million population are suffering chronically from hepatitis-B virus (HBV), and 170 million population are infected with hepatitis-C virus (HCV) (Lok et al., 2001; Daniel, 2009). The recent statistical analysis suggests that the global burden of hepatic complications has increased with time and has shown a huge impact on the overall population throughout the world. It is assumed that compensated cirrhosis and hepatic cancer will reach more than 80% in the year 2020. Besides the viral infections, increased ratio of alcohol consumption and obesity globally, it can be expected that the load of hepatic diseases associated with alcohol and nonalcoholic diseases will be increased twofold. Likewise, those persons having chronic hepatic diseases are more prone to develop severe infections like HIV infection and hepatic carcinoma (WHO, 2002).

3.3 PHYSIOLOGICAL ROLE OF LIVER

The liver is the center of metabolic homeostasis (Fig. 3.1). Some of its major functions are described below.

3.3.1 CARBOHYDRATE METABOLISM

The liver promotes the processing of food particles that are absorbed from the digestive tract after some time. It converts glucose into glycogen which is a storage form of carbohydrate. This glycogen can be reconverted into glucose if there is any demand of energy to operate cellular processes (Guyton and Hall, 2006).

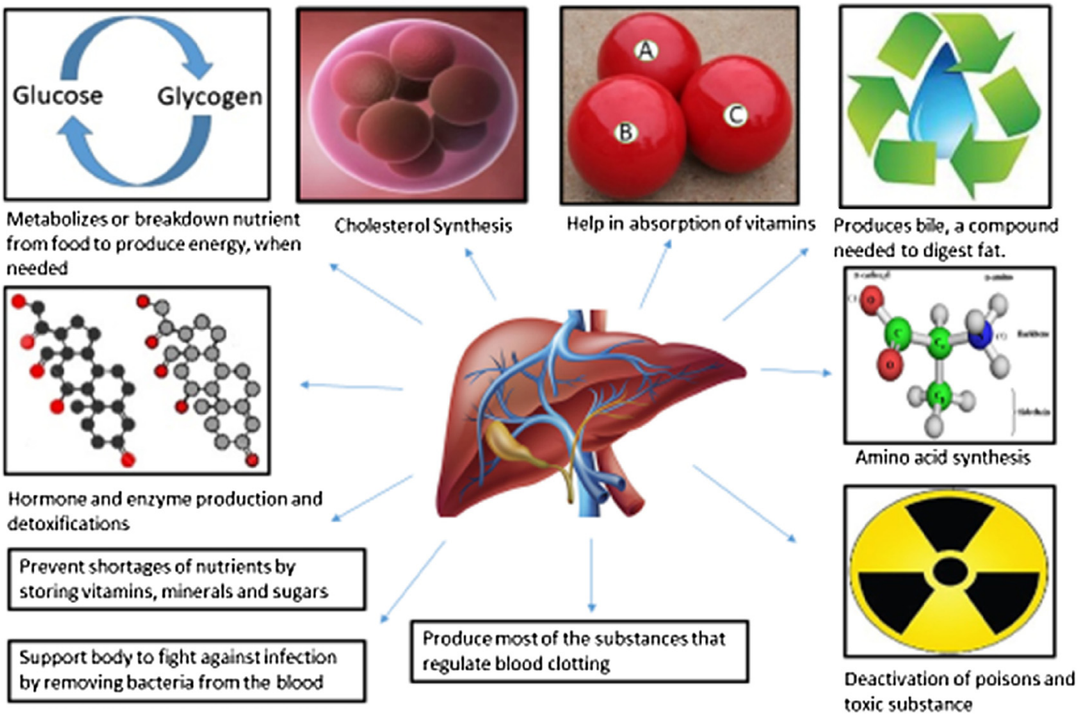

FIGURE 3.1

Physiological roles of liver.

3.3.2 FAT METABOLISM

In the liver, fatty acids oxidation, as well as formation of lipoproteins, cholesterol, and phospholipids are conducted. Fatty acids are synthesized during lipid digestion and are used in the formation of cholesterol and other substances. The liver also play a major role in the conversion of certain amino acids into others (Guyton and Hall, 2006).

3.3.3 PROTEIN METABOLISM AND STORAGE OF VITAMINS

The liver synthesizes blood-clotting protein and plasma proteins. If there are any irregularities in the liver, it can take more time in clot formation in comparison to the normal process. Albumin, another plasma protein formed by the hepatic cells, can bind with various water insoluble elements and promote osmotic pressure. The phagocytic cells of the liver produce acute-phase proteins in response to microbes. These proteins performed the inflammation process, tissue repair, and immune cells activation activities (Reichen, 1999). The liver also acts as a storage organ of certain micro and macromolecules. It stores vitamins including vitamins A, D, K, B12, fats, iron, and copper (Adi and Alturkmani, 2013; Sherwood, 1997). It also accumulates enough glucose in the form of glycogen to provide energy.

3.3.4 NEUTRALIZATION OF TOXIC SUBSTANCE

The liver plays a critical task in eliminating body wastes, toxins, hormones, drugs, and other toxic xenobiotic compounds. These compounds are mixed with blood either by metabolic product formed within the body or in the form of drugs or other foreign complexities. Hepatic enzymes modify most of the toxins for their proper elimination through the urine. Breakdown of hemoglobin leads to the formation of bilirubin and is excreted in the form of bile through the liver (Adi and Alturkmani, 2013). Whereas, hepatic macrophages remove damaged red blood cells from the blood.

3.4 MODE OF HEPATOTOXICITY

Xenobiotics or chemical agents induced toxicity on the liver may contain various mechanisms of cytolethality (Muriel and Rivera Espinoza, 2008; Sharma et al., 2019). Such mechanisms either have a direct impact on cellular organelles including nucleus, endoplasmic reticulum, microtubules, the cytoskeleton, and mitochondria or they have an indirect effect through the stimulation and prevention of signaling kinases, transcription factors, and gene-expression profiles. The resultant intracellular imbalance may lead to cell death caused by either cell necrosis or nuclear disassembly (apoptosis) or by swelling and cell lysis. Some of them are discussed in the subsequent sections.

3.4.1 REACTIVE METABOLITES FORMATION

Many liver toxins such as viz. carbon tetrachloride (Boll et al., 2001), amodiaquine (Zhu et al., 2014), acetaminophen (Piver et al., 2001), isoniazid (Kumar et al., 2013), Metanil yellow (Sharma et al., 2018) are metabolically converted into reactive toxic metabolites and then bind covalently to the major cellular components thus inactivating major cellular functions (Muriel and Rivera Espinoza, 2008). Glutathione provides potential detoxification mechanism for most electrophilic reactive metabolites. Oxidative stress, alkylating agents, and excess substrates for conjugation can trigger the reduction of glutathione thus rendering cells to become more prone to the chemicals' toxicity (Faghihzadeh et al., 2014). The reactive metabolites may also damage hepatic proteins, leading to an immune competent and immune-mediated injury.

3.4.2 LIPID PEROXIDATION AND REDOX CYCLING

Lipid peroxidation is the most common factor associated with cell death due to the overproduction of free radicals which is caused by alteration in the intracellular pro-oxidant to antioxidant equilibrium in favor of pro-oxidants (Sharma et al., 2017). Lipid peroxy radicals increases cell membrane permeability, membrane proteins inactivation, decreases cell membrane fluidity, and loss of mitochondrial membranes polarity. Metal ions contribute in redox cycling while cycling of oxidized and reduced forms of a toxicant lead to the production of free radicals which suppress glutathione activity by oxidizing critical protein sulfydryl groups participated in enzymatic or cellular regulation or can induce lipid peroxidation. Intake of excessive ethanol triggers free-radical production, lipid peroxidation, and glutathione reduction (Saukkonen et al., 2006). Hydrocarbons linked with halogen, hydroperoxides, sodium vanadate, iodoacetamide, cadmium, acrylonitrile, and chloroacetamide are also suggested to exhibit hepatotoxicity because of lipid peroxidation.

3.4.3 EFFECT OF CHEMICAL AGENTS ON MAJOR CELLULAR SYSTEMS

Toxic substance can directly attack certain major cellular components like plasma membrane, mitochondria, endoplasmic reticulum, nucleus, and lysosomes, and thus alter their properties. Hepatic toxins act as an uncoupling agent in mitochondrial electron transport process or directly as an inhibitor (Monshouwer et al., 1995). Covalent binding of intracellular proteins to drug reduces adenosine triphosphate (ATP) levels which in turn enhance actin protein disruption and integrity of cell membrane. Phalloidin, a toxin derived from mushroom increases plasma membrane permeability by binding with actin and thus damaging the cytoskeleton (Wieland and Faulstich, 1978). Toxins such as chenodeoxycholate, phenothiazines, chlorpromazine. and erythromycin show surfactant property in plasma membrane (Seeman, 1972).

3.4.4 MODIFICATION IN CALCIUM HOMEOSTASIS

Calcium governs an inclusive range of essential physiological functions. Cytosolic-free calcium is present in a steadily lower concentration. The concentration of intracellular calcium ion is nearly 10^{-7}M whereas it is 10^{-3}M in the extracellular fluid. Hepatotoxicity induced by a chemical may be associated with calcium homeostasis modifications (Pounds and Rosen, 1988; Nicotera et al., 1990). Nonspecific increase in permeability of the plasma membrane, mitochondrial membrane, and SER membrane leads to calcium homeostasis alterations by upregulating intracellular calcium ion concentration. NADPH, a cofactor associated with calcium pump, may also disrupt calcium homeostasis when present in reduced concentration. Disturbance in calcium homeostasis leads to activation of several membrane destructive enzymes viz. proteases, ATPases, endonucleases, and phospholipases which act upon ATP synthesis, mitochondrial metabolism, and microfilament's structures. Cadmium, acetaminophen, iron, quinines, and peroxides are some of them that show such a type of mechanism on liver injury.

3.4.5 DRUG-INDUCED HEPATOTOXICITY

Drug-induced hepatic anomalies are the primary reason of acute liver failure worldwide (Deng et al., 2009; Kaplowitz, 2004; Lee, 2003). Nearly all drugs are considered as foreign materials to the body and thus are metabolized or eliminated by a wide range of biochemical transformations such as decline of fat solubility and change of biological properties (Fig. 3.2) (Tostmann et al., 2008). Adverse drug reactions can be classified into type-A or type-B reactions. Type-A reactions are dose-dependent and occur in a fairly steady time period. Individuals vulnerable to type-A reactions, usually result in direct toxicity of the native form drug or its metabolites (Pham et al., 1997), for example, acetaminophen-induced hepatotoxicity (Amar and Schiff, 2007) or phenytoin- induced hepatotoxicity (Shear and Spielberg, 1988). Type-B reactions are not directly associated with the pharmacological action of drug (Senior, 2008). It arises in a small population of individuals and persisst at doses that do not lead to toxicity. Additionally, they have not been reproducible in mammalian models (Kaplowitz, 2004), for example, troglitazone-induced hepatotoxicity (Deng et al., 2009) or isoniazid-induced hepatotoxicity (Kumar et al., 2013). Some of the drugs develop toxicity and possess potential adverse effects.

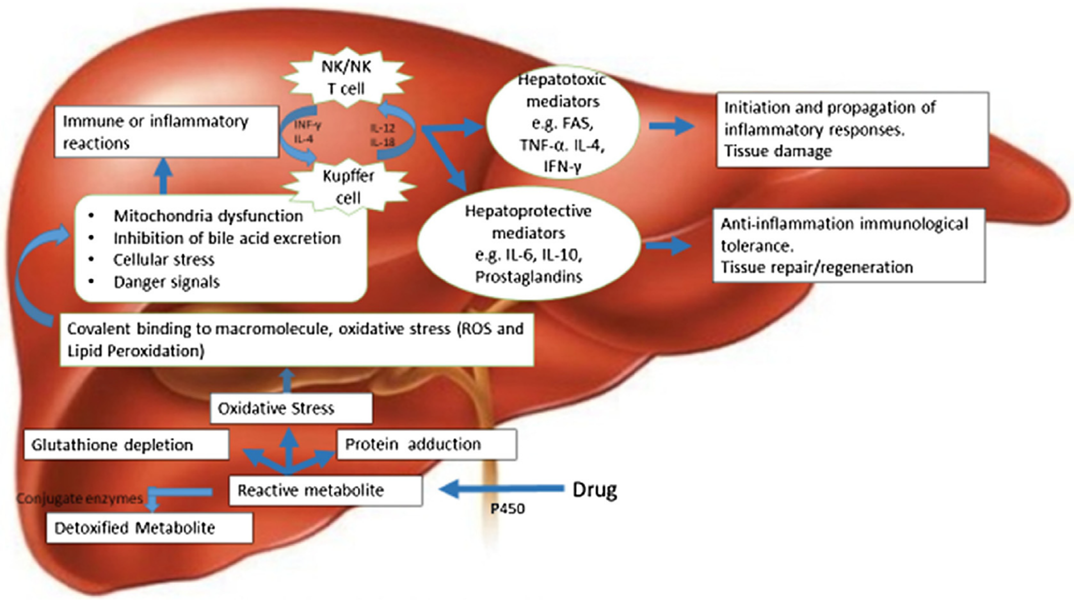

FIGURE 3.2

Metabolism of drug and its associated complications in liver.

3.4.5.1 Antitubercular drugs

Antitubercular drug-induced hepatic injury (ATDH) is a severe problem and a major cause of treatment disruption and alteration in routine treatment during tuberculosis (TB) management course (Khalili et al., 2009). ATDH leads to a high rate of morbidity and mortality wordwide. Elevations of serum transaminases are a common sign in the case of the antituberculosis treatment. However, hepatic damage can be dangerous when not identified at the initial stage or medication is interrupted during treatment. Drugs including isoniazid, paracetamol,l and pyrazinamide have been suggested to be highly toxic to the liver. The chance of antitubercular drug associated toxicity has been increased by various factors. Some of them are chronic viral infection, preexisting chronic liver disease, nutritional status, high intake of alcohol, older age, advanced stage of TB, contaminated food, overuse of hepatotoxic drugs, and inappropriate dose of drugs (Tostmann et al., 2008; Khalili et al., 2009). Detoxification of drugs and its metabolites are interrelated which depends on the potential of hepatic enzymes (Chang and Schaino, 2007). Most antitubercular drugs are lipid soluble and are transformed into aqueous compounds by phase I and II biotransformation enzymes in liver. Toxicity arises by isoniazid which is considered idiosyncratic, that is, reactive toxic metabolites rather than the native form of drug which are primarily responsible for toxicity. A rapid onset of toxicity was detected in the rabbit model after treatment with isoniazid at 3-hour intervals for 2 days (Sarich et al., 1996). Some animal models showed elevated serum transaminases levels for a maximum duration of 36 hours during hepatic necrosis. The mechanism of action of hepatotoxicity associated to rifampicin is not fully known and hence there is no proof for the presence of toxic metabolite of this drug (Westphal et al., 1994). Administration of rifampicin along with isoniazid increases the risk of developing hepatotoxicity (Steele et al., 1991). Rifampicin promotes isoniazid hydrolase which in turn increases hydrazine synthesis and thus leads to toxicity in combination. Another antitubercular drug pyrazinamide, inhibits the action of CYP450 isoenzymes (Maffei and Carini, 1980). But during the investigation of mammalian hepatic microsomes, it was found that it possesses a noninhibitory effect on the CYP450 isoenzymes (Nishimura et al., 2004).

3.4.5.2 Anticoagulants drug

Oral anticoagulants drugs like phenprocoumon, ximelagatran, acenocoumarin, warfarin, heparin, and enoxaparin are frequently used for avoidance of clot formation in blood vessel and stroke (Arora and Goldhaber, 2006). Anticoagulant-induced hepatic abnormalities have been seen to be connected with asymptomatic increase of serum transaminases, clinically important hepatitis, and serious hepatic failure. Ximelagatran, warfarin, and dabigatran upregulates alkaline phosphatase (ALP); whereas, jaundice was identified with ximelagatran and warfarin (Arora and Goldhaber, 2006). Phenprocoumon-related hepatic injury associated with direct injury of hepatocytes by reactive metabolites resulted into amplified antigenicity and a consequent immune-allergic reaction. It also involves CYP450 enzymes that cause decline of ionic gradients, reduction of ATP levels, cell edema, and damage (Schimanski et al., 2004). Heparin is involved in direct toxicity, immune-mediated hypersensitivity, and membrane modification reaction in liver (Carlson et al., 2001).

3.4.5.3 Anticonvulsants or antiepileptic drugs

Anticonvulsant drugs may also be associated with hepatotoxicity. Clonazepam, primidone, chloral hydrate, sultiam, and diazepam are not causes of severe hepatic abnormalities. Sodium valproate is a prominent anticonvulsant with a lower rate of hepatotoxicity (Green, 1984). It is transformed into valproyl adenosine monophosphate and valproyl coenzyme A in the mitochondrial matrix. The valproate induction promotes the diminution of coenzyme A, thus affecting the intra-mitochondrial storage of this cofactor and impairs mitochondrial enzymes which are involved in ß-oxidation of fatty acids (Murray et al., 2008). Phenobarbital is hardly recognized to develop hepatic abnormalities with hepatocellular and cholestatic liver injury and also hypersensitivity reaction. Individuals who take phenytoin generally have threefold elevated levels of transaminase as compared to normal values (ULN). Intake of felbamate was expressively lowered due to its relation with aplastic anemia and sometimes hepatic injury (Deng et al., 2009).

3.4.5.4 Antihyperlipidemic drugs

Ezetimibe that lowers the cholesterol by decreasing its absorption at the small intestinal brush border cell, sometimes leads to hepatic injury such as acute autoimmune hepatitis and chronic cholestatic hepatitis (Stolk et al., 2006). Fenofbrate activates an autoimmune hepatitis response specially when taken with statin medications (Alsheikh-Ali et al., 2004). Atorvastatin and lovastatin-mediated hepatic abnormalities have been connected with a diversified action of liver damaged, normally occurring months after the beginning of the treatment (Bhardwaj and Chalasani, 2007). The mode of injury from antihyperlipidemic is classically hepatocellular or mixed in nature with unusual cases of pure cholestatic hepatitis (Chitturi and George, 2002; Parra and Reddy, 2003). Provastatin has been reported to develop acute intrahepatic cholestasis (Hartleb et al., 1999). Simvastatin-induced hepatic abnormalities are supposed to arise due to drug-to-drug interactions (Ricaurte et al., 2006).

3.4.5.5 Antiretroviral and antimalarial drugs

Liver destructive action of antiretroviral drugs revealed that three different classes, including protease inhibitors, nucleoside reverse transcriptase inhibitors (NRTIs), and nonnucleoside reverse transcriptase inhibitors (NNRTIs) are mainly responsible for toxicity (Brind, 2007). They develop hepatic abnormalities by various mechanisms such as mitochondrial impairment by nucleoside equivalents like stavudine and didanosine, hypersensitivity reactions by nevirapine, efavirenz, or abacavir, direct hepatic harm by using maximum doses of ritonavir and immune reconstitution process, mainly in rigorously immunosuppressed individuals with primary chronic HBV. Nucleoside analogs (NRTIs), particularly zerit, videx, and retrovir are linked with hepatic steatosis and lactic acidosis (Piroth, 2005). Steatohepatitis promotes the advancement of hepatic fibrosis in a person with severe HCV infection. NNRTIs, especially viramune, are related to hepatic necrosis and hepatitis. It is assumed that for the liver, nevirapine is more toxic in comparison to efavirenz (Van Leth et al., 2004). The underlying prolonged HCV infection promotes the risk of elevation of hepatic enzyme. Inhibitors of protease have been concerned with periods of hepatotoxicity, with fosamprenavir/low-dose ritonavir, lopinavir/low-dose ritonavir, and nelfnavir being less toxic (Sulkowski et al., 2004), and tipranavir/ low-dose ritonavir being more hepatotoxic (Rockstroh et al., 2006). Indeed, a person suffering with chronic HCV is at risk of enzymatic alterations following exposure to most antiretroviral medications.

Amodiaquine, an antimalarial drug can develop liver disorder in human beings by transforming into reactive metabolite, iminoquinone by hepatic peroxidases and microsomes (Shimizu et al., 2009). The reactive metabolites can permanently bind to proteins which develop toxicity by altering the cellular function. These types of individuals have been observed to have antidrug IgG antibodies (Clarke et al., 1991). Amodiaquine can induce immune response in a mammalian model similar to that in humans, but it is not enough to result in clinical toxicity (Clarke et al., 1990).

3.4.5.6 Chemotherapeutic drug, glucocorticoids, and anabolic androgenic steroids

Chemotherapeutic medications use poisonous chemicals or drugs like antimetabolites, antitumor antibiotics, platinum, tyrosine kinase inhibitors, alkylating agents, biologic response transformers, and androgens to destroy carcinoma cells (King and Perry, 2001; Perry, 1992). During treatment process, if these toxins accumulate in the body faster than its degradation in the liver, hepatic abnormalities arises (Lim et al., 2010). Chemotherapeutic drugs alone or in association with other drugs may develop liver toxicity or hypersensitivity reactions (King and Perry, 2001).

Glucocorticoids enhance storage of glycogen in hepatocytes. Administration of steroidal drugs is an unusual cause of inflamed liver in children, while prolonged use of steroids leads to developing steatosis in human beings (Iancu et al., 1986; Alpers et al., 1982). Anabolic androgenic steroids, generally referred to as nutritional supplements, cause severe hepatotoxicity (Saukkonen et al., 2006; Kafrouni et al., 2007).

3.4.5.7 Nonsteroidal antiinflammatory drugs

Nonsteroidal antiinflammatory drugs (NSAID) like acetylsalicylic acid elevate hepatic enzymes and develop acute cytolytic, cholestatic, or mixed hepatitis conditions. Serum transaminases and ALP are specific liver markers that are used as an early warning sign for hepatic abnormalities. In critical cases, there may be supplementary signs and symptoms of anorexia, vomiting, nausea, abdominal pain, weakness, and jaundice along with increased prothrombin time and bilirubin level (Manov et al., 2006). In that case, NSAID both dose-dependent and idiosyncratic-based mechanism have been involved. Phenylbutazone and aspirin have been associated with intrinsic hepatotoxicity. Whereas, Ibuprofen, piroxicam, sulindac, diclofenac, indomethacin, and phenylbutazone are concerned with idiosyncratic reaction; the biochemical and clinical significance of diclofenac in living beings are concerned with both impairment of ATP production by mitochondria and reactive metabolites formation specifically N, 5-dihydroxydiclofenac, which leads to direct injury.

3.5 PHYTOCHEMICALS

Plants are the major source of medications in folk medicinal systems, nutraceuticals, pharmaceutical intermediates, modern medicines, food supplements, and also act by adding chemical moieties in synthetic medicines (Dewick, 1996). Phytochemicals are categories under several classes such as flavonoids, phenol, carotenoids, alkaloids, tocopherols, terpenoids, polyphenols, saponins, phytosterols, and organosulfur compounds. Phytochemicals within planst generally provide protection against pests, pathogens, predators, and the external environment (Cowan, 1999). There are more than 7500 different polyphenolic compounds are identified in plant kingdom which are medicinally important. Grains, Cereals, fruits, legumes, nuts, vegetables and spices are rich in polyphenols (Opara and Rockway, 2006). A good knowledge of essential and nonessential compounds of plant origin and their actions for biological activities in the term of health and diseases are necessary for accepting nutrition as a tool for health management (Gupta et al., 2017). Natural compounds are widely present in diets and have potential to regulate various metabolic pathways, which protect living beings from various ailments. Broad spectrum in vitro, in vivo, and clinical studies suggested that the natural bioactive compounds effectively govern acute and chronic hepatic complications (Opara and Rockway, 2006).

3.5.1 CURRENT STATUS OF THERAPEUTIC PLANTS

Use of herbal formulation throughout the world tended to decrease in the late 19th to early 20th centuries. The World Health Organization's (WHO, 2001) report revealed that around 60% of the world's population uses traditional medicine, while nearly 80% of the population of developing countries are completely based upon traditional system of medicine (Fransworth, 1994). Currently, a folk herbal medicinal system is used in China, India, Nepal, Bhutan and other developing countries (Pandey, 2007). Medicinal plants are actively used in modern drugs by providing active compounds upon which a new drug is synthesized. Newman and coworker (2003) reported that nearly 75% of antiinfectious drugs and 60% of anticancer drugs approved during 1981–2002, have been derived from natural resources, although 61% of all newly discovered chemical moieties recommended as a drugs during the same period were inspired by natural compounds (Gupta et al., 2005). Hence the search for new medicinal herbs for drug advancement and dietary ingredients have enhanced in the last few years. Botanists, microbiologists, biochemist and pharmacologists are currently discovering medicinal herbs for phytochemicals and active components that would be useful for management of various diseases (Mishra et al., 2013).

3.5.2 HEPATOPROTECTIVE PROPERTIES OF PHYTOCHEMICALS AND THEIR MODE OF ACTION

Bioactive compounds are a natural source of antioxidants and antiinflammatory biomolecules, which are actively used in amelioration of hepatic injury. Generally they are profusely present in plants such as phenol, flavonoids, alkaloids, polyphenols, terpenoids, glycosides, and saponins. Several in vivo, in vitro, and ex-vivo studies have suggested that bioactive components possess immunomodulatory, antioxidant, antimicrobial, antimutagenic, hypoglycemic, hypolipidemic, antiinflammatory, gastroprotective, anticancer, and chemopreventive activities (Kumar and Pandey, 2013). Bioactive components suppress the antagonistic effects of drugs/chemicals and their metabolites. These compounds up-egulate intracellular signaling pathways and trigger Nrf2 molecule through initiation of the P1K3/AKt and extracellular-signal-regulated kinase (ERK), and thereby it operates various transcriptional factors but alternately it adversely regulates SP/NR1 signaling pathways. In general, bioactive compounds perform a critical role in neutralization of oxidative stress induced by drugs/chemicals and normalizes intracellular enzymes, protects cells from toxicity, and detoxifies various toxic compounds from the cell.

3.5.2.1 Glycyrrhizin

Glycyrrhizin, an active constituent of *Glycyrrhiza uralensis*, is used as a folk medicine in hepatitis. Fujisawa et al. (2000) suggested that it possesses significant antiinflammatory properties in hepatic and other cellular organs. Glycyrrhizin regulates the cytolytic activity of complement system by stimulating both alternative and classical pathways. Additional studies demonstrated that glycyrrhizin reduces the lytic process where membrane attack complex is formed. In vivo analysis has shown that glycyrrhizin can prevent hepatic inflammation by increasing the secretion of interleukin-10 (IL-10) in concanavalin-A (Con A)-induced hepatitis in animal model. Advance studies on mechanism disclose that glycyrrhizin can alter the mouse dendritic cells properties with autoimmune hepatitis and enhance antiinflammatory cytokines' production (Abe et al., 2003). A European clinical study revealed the medicinal properties of glycyrrhizin against viral hepatitis. Sixty-nine out of seventy-two individuals completed treatment, in the untreated group, there was no change observed in ALT level. Whereas, three and six-time glycyrrhizin-treated individuals recovered ALT enzyme toward normal level at the end of treatment. In the six-time treated group, 20% (3 of 15) of individuals attained normal ALT levels and the three-time treatment group, the result was about 10% (4 of 41). No significant side effects have been observed during the study (Van Rossum et al., 2001).

3.5.2.2 Resveratrol

Resveratrol (3,5,4′-trans-trihydroxystilbene), a polyphenolic compound, has been isolated from grapes, red wine, peanuts, berries, and other natural sources. It possesses antioxidant, antiinflammatory, anticarcinogenic, and antifibrogenic properties (Mates et al., 2011; Chvez et al., 2008). During the process of metabolic breakdown it is converted into resveratrol sulfate and a small fraction of resveratrol glucuronide (Yu et al., 2002). When resveratrol was given in the form of aglycone or glucuronide, it promotes enterohepatic circulation (Marier et al., 2002). Williams et al. (2009) on their study reported that high-purity transresveratrol at several times of exposure and doses did not show any kind of toxicity. Furthermore, they did not find any adverse effect on 700 mg/day for 90 days as the higher dose and time of exposure. Cheng et al. (2012) revealed that resveratrol can trigger ERK signaling pathway, which in turn enhanced the activation and translocation of Nrf2 to the nucleus, thus upregulatesd HO-1 and glyoxalase expression. Another study also supported that resveratrol by promoting the translocation of Nrf2 to the nucleus, provided an alternative pathway for protection from oxidative stress (Bagul et al., 2012). Resveratrol lowers the acetylation of peroxisome proliferator-activated receptor gamma coactivator 1-a and upregulates its activity by promoting protein deacetylase sirtuin 1, thus improving mitochondrial function and protecting against metabolic abnormalities (Lagouge et al., 2006). In an experiment designed by Crowell et al. (2004) suggested that resveratrol at the highest dose (3000 mg/day for 4 weeks) leads to renal abnormalities with a decrease body weight, food intake, and other tissue-specific markers lesions.

3.5.2.3 Tragopogon porrifolius

Tragopogon porrifolius (family Asteraceae) is well known for its edible shoot and root (Mroueh et al., 2011). The nutritive importance of this plant is related to vitamins, polyphenols, fructo-oligosaccharides, and monounsaturated and essential fatty acids, possesses probiotic activity on the intestinal microflora. Hentriacontane, hexadecanoic acid, hexahydrofarnesylacetone, and 4-vinyl guaiacol, are some of the most abundant bioactive compounds present in *T. porrifolius* (Konopinski, 2009; Formisano et al., 2010). It has shown hepatogenic/hepatoprotective property in various hepatic abnormalities induced by diverse toxic agents including drugs, chemicals, pollutants, and parasites, bacteria, or viral infections. These therapeutic attributes of plants are associated with the polyphenolic compounds. Antioxidant efficacy of methanolic extract of *T. porrifolius* aerial part against $CC1_4$-mediated hepatic injury in rodents showed enhanced activity of hepatic antioxidant enzymes in a dose-dependent manner. Nearly 250 mg/kg concentration of extract increases the activity of SOD, CAT, and GST. Constituently, in $CC1_4$-mediated hepatic injury, the plant extract restored the activity of AST, ALT, and LDH toward normal level (Govind, 2011). Whereas, *T. porrifolius* stem aqueous extract during inflammation, lipemia, gastric ulcer, oxidative stress, glycemia, and hepatotoxicity showed significant reduction in the levels of serum glucose, cholesterol, triglyceride, and hepatic enzymes. In addition, *T. porrifolius* also exhibited potent radical scavenging activity (Zeeni et al., 2014).

3.5.2.4 Silymarin

Silymarin is a combination of flavonolignans extracted from *Silybum marianum* (milk thistle seeds). Major constituents of silymarin comprises silibinin, silidianin, silicristin and isosilibinin. Silibinin, an active compound of silymarin has been extensively used as hepatoprotective and anticancerous agent. Bonifaz et al. (2009) studied the antiviral efficacy of silymarin and suggested that it can suppress protein expression and reduce multiplication of HCV core mRNA. The antiviral activity of silymarin differs because of interferon which suppresses the JAK/STAT signaling pathway. It prevents the access and transmission of HCV by averting microsomal triglyceride transfer protein activity and decreases production of apolipoprotein B. Mayer and coworkers (2005) on their study reported therapeutic efficacy of silymarin against HBV and HCV. They suggested that in hepatitis patients silymarin can decrease serum transaminases levesl, although the serum viral content and histology have not been altered by silymarin. A clinical study on the immunomodulatory activity of silymarin against hepatitis-C patient revealed that it can prevent inflammation during in vivo and in vitro analysis. It also decreases the production and proliferation of T-cell by lowering pro-inflammatory cytokine with enhancing antiinflammatory IL-10 (Adeyemo et al., 2013). Silibinin also showed the inhibitory effect on HCV by suppressing clathrin-dependent trafficking. It promotes assembly of clathrin-coated pits and vesicles in hepatocytes and terminates the clathrin associated endocytic pathway by disturbing the absorption and trafficking of transferrin (Blaising et al., 2013).

3.5.2.5 Phyllanthus niruri

Phyllanthus niruri is a subtropical plant, widely spread throughout the world. It is used as folk medicine against chronic hepatitis, intestinal infections, and kidney disease (Liu et al., 2014; Ibrahim et al., 2013). Quercetin rhamnoside, quercetin glucoside, gallic acid, and geranin are some major bioactive compounds identified in *P. niruri* which possess hepatoprotective, antibacterial, and antiviral properties (Amin et al., 2012). In the past few years, studies were designed to disclose the antiviral property and safety of *P. niruri* by chemical and biochemical experiments. Lam et al. (2006) reported that *P. niruri* ethanolic fraction could suppress HBsAg production with downregulating the expression of HBsAg mRNA which shows that its mechanism may be associated with the upregulation of annexin A7 protein.

3.5.2.6 Salvia miltiorrhiza

Salvia miltiorrhiza is a folk Chinese medicine used for the management of cardiovascular and cerebrovascular disorder. Recent studies have proposed that it can upsurge blood circulation in hepatocytes to suppress the potential destruction by removing toxic compounds from the liver. The active compound of *S. miltiorrhiza* can also possess hepatoprotective efficacy against CCl_4 associated hepatotoxicity. It shows hepatic protection due to down-regulation of NF-kB and p38 signaling pathways in hepatic Kupffer cells (Yue et al., 2014). Treatment of high iron comprising mice with *S. miltiorrhiza* will recover normal morphology of liver, reducing iron accumulation and down-regulating the expression of collagens (type I and III), TGF-ß mRNA and triggering the expression of matrix metalloprotein-9 mRNA in the hepatocytes. Moreover, *S. miltiorrhiza* can also reduce malondialdehyde (MDA) content and increase SOD activity and glutathione (GSH) content while down-regulating tumor necrosis factor (TNF)-α, IL-1a, and caspase-3 expression (Zhang et al., 2013a,b).

3.5.2.7 Gallic acid

Gallic acid, a trihydroxybenzoic acid is a major constituent of *Punica granatum* pericarp. Extensive studies have proven that it has strong antiobesity and antioxidant activities. Chao and cowerker (2014) revealed that it recovers the glucose and lipid metastasis associated anomalies in NAFLD mice. It shows hepatoprotective efficacy by regulating amino acids, choline, lipid, and glucose metabolism. Histological study of high fat diet-induced NAFLD rat indicated that the lipid droplets were remarkably reduced in gallic acidtreated group than control group. Cholesterol level and triacylglycerol were also found to be expressively lesser after gallic acid treatment. Moreover, gallic acid administration reduced oxidized glutathione (GSSG) content and amplified the antioxidative enzyme level. These observations suggested that gallic acid can suppress NAFLD associated oxidative stress, dyslipidaemia and hepatosteatosis in rats (Hsu and Yen, 2007).

3.5.2.8 Baicalin

Baicalin ($C_{21}H_{18}O_{11}$) is an active constituent of *Scutellaria baicalensis* and capable of protecting the liver from oxidative damage by promoting fatty acid binding protein expression and activity of intracellular enzymes (Ai et al., 2011). In vivo studies revealed that it contains antiinflammatory, antioxidant, and antiapoptotic properties, and supports its hepatoprotective efficacy during hepatic ischemia/reperfusion induced aberrations (Kim et al., 2010). Wan et al. observed that a diet having baicalin (0.25% and 1%) possesses hepatoprotective effects against excessive iron intake associated with hepatotoxicity in mouse during a 50 day treatment. The mechanism behind this may be an iron chelation and antioxidant property of baicalin (Zhao et al., 2005). Baicalein, ($C_{15}H_{10}O_5$), another bioactive component extracted from *S. baicalensis*, also established agent against hepatic abnormalities. A recent study demonstrated that baicalein promotes hepatic cells regeneration by modifying IL-6 and TNF-a facilitated signaling pathways (Huang et al., 2012). Moreover, molecular mechanisms of apoptosis in hepatocytes were shown to have a protective effect on mitochondria, prevention of the cytochrome C oozing, decrease of Bax/Bcl-2 ratio, and reduction of transcription factor phosphorylation (NF-kB, JNK and ERK) (Wu et al., 2010).

3.5.2.9 Prangos ferulacea

Prangos ferulacea is an Iranian herbal medicine commonly used in gastrointestinal abnormalities. In addition to Iran, other species of this genus are found in Caucasia, Central Asia, and East Europe to Turkey. It has been used in traditional medicine such as emollient, carminative,e and tonic for gastrointestinal complications, sedative, antibacterial, antiviral, antifungal, antihelminthic, antiinflamatory, and antiflatulent. Flavonoids, alkaloids, sesquiterpenes, saponins, coumarines, tannins, monoterpenes, and terpenoids are essential constituents identified in *P. ferulacea*. It was found that the oils isolated from fruits and leaves were rich in monoterpenes, mainly α- and β-pinene, while some of them showed antioxidant activity against oxidative damages. In a study, the antioxidant efficacy of *P. ferulacea* is described to be higher in comparison to α-tocopherol. Hydroalcoholic extract of *P. ferulacea* modified hepatic physiology and the activities of serum AST and ALT. In alloxan-induced diabetes in rats, the serum ALT and AST levels increased significantly (Massumi et al., 2007; Akhgar et al., 2011). Moreover, hepatic necrosis, vacuole formation in cytoplasm and lymphocytic inflammation was also observed. Treatment of diabetic rats by root extract exhibited remarkable decrease in enzymes. Hydroalcoholic extract of root has also shown alterations in aminotransferases activity and prevents the histopathological changes in the liver associated with alloxan-induced diabetes (Kafash Farkhad et al., 2012).

3.5.2.10 Dioscin

Dioscin is a steroidal saponin found exclusively in *Dioscorea opposita* and has been medicinally accepted for their protective activity in fatty liver. Recent studies validated that it can slow down the process of lipid storage in the liver and hence reduced body weight, enhanced energy expenditure with more oxygen consumption, and decreased serum AST, ALT levels. Another study on mechanisms explored that dioscin combats oxidative damage, prevents inflammation, prevents triglyceride and cholesterol synthesis, reduces mitogen-activated protein kinase phosphorylation levels, triggers β-oxidation of fatty acid, and enhances autophagy to improve fatty liver conditions. These outcomes suggested that dioscin may be a suitable compound for obesity and NAFLD treatment (Liu et al., 2015). Various other studies also revealed its tremendous protective property against alcoholic fatty liver by countering alcohol-induced oxidative stress, inflammatory cytokine production, apoptosis, mitochondrial function, and in vivo hepatic steatosis (Xu et al., 2014).

3.5.2.11 Schisandra chinensis

Recent therapeutic studies on *Schisandra chinensis* reported that it may control the body's humoral balance, trigger immune responses, and possess anti-HBV properties (Yim et al., 2009). Xue et al. (2015a,b) isolated and studied seven new lignans and a schischinone, along with some known lignans from its fruit. They observed that two lignans showed remarkable anti-HBV property by preventing HBV DNA replication with less significant hepatotoxicity. Schisandrin B, a dibenzocyclooctadiene compound present in *S. chinensis* fruit possesses healing property against hepatitis. The antiinflammatory activity of Schisandrin B was studied by Checker et al. (2012) and disclosed that it stimulated translocation of nuclear factor erythroid-derived factor 2-related factor (Nrf2) and enhanced the transcription of HO-1. Schisandrin B also prevented the translocation of NF-kB in stimulated lymphocytes and down-regulated the expression of downstream inflammation related genes.

3.5.2.12 Periplocoside A

Preplacoside A (PSA), a glycoside of *Preplacoside sepium* is exclusively used in rheumatoid arthritis as traditional medicine. It was found that pretreatment with PSA has remarkably improved the hepatic damage, secretion of interleukin-4, interferon-γ, and ALT levels were intensely reduced, which further repressed the hepatic necrosis and normalized the hepatic function in vivo. In vitro evaluation revealed that PSA decreased cytokines secretion, which is necessary for inflammatory response viz., IL-4, and IFN-γ produced by natural killer T-cells upon stimulation with anti-CD3 mAb or a-galactosylceramide. Moreover, no significant toxicity of PSA has been observed during in vitro and in vivo analysis. These findings supported that PSA may have a therapeutic potential for autoimmune-related hepatitis management in humans (Wan et al., 2008; Zhang et al., 2009).

3.5.2.13 Astragalus membranaceus

Astragalus membranaceus stimulates hepatic metabolic processes, enhances lung activity and gastrointestinal tract, stimulates wound restoration, and provides relief from fatigue. Saponins and flavonoids including calycosin, formononetin, astragaloside, and calycosin-7-Obeta-D-glucoside are the main bioactive compounds present in *A. membranaceus* (Cai et al., 2015; Yu et al., 2014). Recent studies proposed that it could alter immunologic system, prevent virus development, and prevent cancer cell proliferation (Gao et al., 2014; Liu, 1991). Some researchers explored the beneficial properties of *A. membranaceus* in an individual with viral hepatitis. Furthermore, in vivo analysis suggested that it prevents DNA replication in duck HBV. *A. membranaceus* treatment also declines pathological alterations in duck HBV than the untreated group. Advance molecular studies have supported that active components of *A. membranaceus* suppress HBV activities which may be due to the presence of triterpenoid saponin (Wang et al., 2009).

3.5.2.14 Argimonia eupatoria

Argimonia eupatoria is employed for the treatment of various disorders such as inflammation. Its aqueous extract enrich numerous phenolic compounds while ethyl acetate extract has shown to possess antioxidant potential with lower toxicity (Correia et al., 2007). *A. eupatoria* is rich in phenolic, coumarins, tannins, terpenoids, and flavonoidal compounds viz. coumaric acid, quercitrin, chlorogenic acid, gallic acid, and protocatechuic acid (Giao et al., 2009). The hepatoprotective activity of *A. eupatoria* was evaluated against diethylnitrosamine and $CC1_4$-mediated hepato-carcinogenesis, whereas, aqueous extract was tested against ethanol-induced hepatic injury. Oral administration of *A. eupatoria* extract at concentration 10, 30, 100, and 300 mg/kg against chronic consumption of ethanol pronouncedly normalized serum aminotransferase activities and pro-inflammatory cytokines. CYP450 activity and lipid peroxidation were elevated with reduced glutathione concentration during ethanol consumption. Plant extract also mitigated oxidative stress and toll-like receptor-related inflammatory signaling (Yoon et al., 2012).

3.5.2.15 Panax notoginseng

Panax notoginseng is a member of genus *Panax*. It has been found that *P. notoginseng* extracts reduced ethanol-induced lipid accumulation by suppressing the formation of MDA, glutathione (GSH) and ROS, downregulating IL-6 and TNF-α levels, along with promoting SOD activity in liver and prevent cytochrome P450 2E1 (CYP2E1) induction (Ding et al., 2015; Yang et al., 2009). In vivo analysis of *P. notoginseng* against NAFLD individuals revealed that saponins present in it can release oxidative stress and insulin resistance. Concentration of MDA, hydroxy radical ($OH^{\cdot}$), and pro-inflammatory cytokine (TNF-α) decreases after *P. notoginseng* administration. Whereas, total antioxidant capacity and SOD activity were improved with upgraded insulin resistance.

3.5.2.16 Puerarin

Puerarin is a form of isoflavones obtained from *Pueraria lobata*. It stimulates hepatic metabolic function and lowers serum ALT, AST, and total-bilirubin level with lesser ECM contents while increases the amount of albumin and total protein in hepatic fibrosis. In vivo study suggested that puerarin could prevent the CCl_4-induced liver fibrosis. Its mechanisms may be related with the alteration of peroxisome proliferator-activated receptor (PPAR)-γ protein expression and PI3K/Akt signaling pathway (Guo et al., 2013). Puerarin associated TNF-α/NF-kB signaling may also promote antifibrosis efficacy in liver. Moreover, puerarin administration resulted in a decline expression of TNF-α and NF-kB, the inflammatory response in liver was regularized and metabolic activities were also enhanced (Li et al., 2013). Puerarin can also induce apoptosis in hepatic stellate cells, which perform a crucial role in hepatic fibrosis. Furthermore, the mechanism associated studies revealed that puerarin in hepatic stellate cells induced apoptosis by down-regulation of bcl-2 mRNA cells (Zhang et al., 2006).

3.5.2.17 Camptothecin

Camptothecin is referred to as a toxic quinoline alkaloid, extensively used as a DNA topoisomerase-I suppressor in cancer treatment. It is typically isolated from *Camptotheca acuminata* stem or bark and worked as an anticancer agent in folk Chinese medicine. Li *et al.* (2011) observed that it can down-regulate SMMC-7721 cell progression by blocking S and G2/M stages of cell cycle and promote mitochondrial associated apoptosis pathway. It can also upregulate TRAIL-mediated apoptosis in HCC cells by suppressing free radicals formation and ERK/p38-dependent DR5. Additional investigation supported HCC cells pretreated with antioxidants or DR5 specific inhibitors of p38 and ERK can suppress camptothecin- TRAIL-related cell apoptosis by preventing DR5 expression (Jayasooriya et al., 2014). In vivo studies supported that the camptothecin in combination with *N*-trimethyl chitosan (CPT-TMC) may increase solubility and enhance potential for hepatic cancer treatment. In liver oncogenic BALB/c mice model, CPT-TMC remarkably reduces cancer progression and lymphatic metastasis along with extended survival time, while no significant side-effect have been observed. Thus, CPT-TMC combination is a nontoxic and effective medicinal compound for hepatic cancer therapy (Zhou et al., 2010).

3.5.2.18 Citrullus lanatus

Citrullus lanatus fruits are edible as a febrifuge when ripe completely or almost putrid. Its fruit is diuretic and useful for the treatment of dropsy and kidney stones. Fruit peel is recommended in diabetes and alcoholic toxicity. *C. lanatus* aqueous extract is a good source of lycopene, glucose, vitamin C, fiber, and β-carotene. Watermelon juice suppresses SOD activity and LDL-cholesterol, while enhances CAT activity and HDL-cholesterol, which could reveal its antioxidant property (Georgina et al., 2011). The majority of cucurbitacin has shown cytoprotective activity against HepG2 cells. Cucurbitacin was suggested to have high efficacy as antifibrosis agent. Administration of $CC1_4$ once in a week for 28 days caused remarkable upsurge of hepatic serum markers viz. AST, ALT and total bilirubin, while there was a reduction in albumin as compared to the untreated group. Furthermore, $CC1_4$ along with ursodeoxycolic acid or watermelon juice appreciably diminished these alterations. After $CC1_4$ administration, the LPO level elevated in kidney, brain, and liver tissues. Meanwhile, ursodeoxycolic and acid watermelon juice treatment inhibited increment in LPO. Outcomes suggested that, watermelon juice protecedt liver from $CC1_4$-induced toxicity probably due to the antioxidant potential and prevention of production of lipid peroxide (Altas et al., 2011).

3.5.2.19 Curcumin

Curcumin is a diphenylheptanoid extracted from turmeric. It is mainly used as food colorant and has also been recommended in various studies for its significant anticancer effects in in vitro and in vivo analysis. Notarbartolo et al. (2005) revealed that when curcumin is used alone or in combination with doxorubicin or cisplatin, it can suppress cancer cell propagation and promote apoptosis in the liver. Further studies suggested that curcumin altered activation of NF-kB and IAP gene expression. For curcumin-resistant hepatic carcinoma cells, a recent analysis has explored that Chk1-associated G2/M cell cycle block may be linked with curcumin resistance and Chk1 could be an effective site for future analysis in retreating resistance and promoting the clinical efficacy of curcumin (Wang et al., 2008).

3.5.2.20 Berberine

Berberine is a quaternary ammonium salt obtained from plant of Berberis, such as *Coptis chinensis*, that has been proposed as a potent natural compound which prevents triglyceride accumulation (Fan et al., 2013). It can promote insulin resistance in NAFLD by upregulating the expression of insulin receptor substrate-2 and altering signaling pathway associated with insulin (Xing et al., 2011). In a study berberine-containing nanoparticles (BBR-SLNs) can release hepatosteatosis in by lowering lipogenesis and enhancing lipolysis in hepatocytes. Additional investigation of mechanisms suggested that these nanoparticles can lower the lipogenic genes expression such as stearoyl-CoA desaturase, sterol regulatory element-binding protein 1c and fatty acid synthase, and upregulate lipolytic gene expression including carnitine palmitoyltransferase-1 (Xue et al., 2015a,b).

3.5.2.21 Hesperidin

Hesperidin ($C_{28}H_{34}O_{15}$) is a glycosidic flavanone present enormously in citrus fruits. It is assumed that hesperidin play an important role in plant defense system. In vitro studies proposed that it act as a good antioxidant and maintain blood vessels' integrity (Farombi et al., 2008). Hesperidin has strong healing property against DNA damage, suppressing the activity of micronuclei damaged by UV-irradiation (Hosseinimehr and Nemati, 2006). It can retrieve the liver against doxorubicin-induced hepatotoxicity by boosting the activities of serum and tissue-specific enzymes. In addition, it has shown a remarkable rise in hepatic glutathione level, glutathione peroxidase, glutathione-S-transferase, and peroxidase activities, and has lowered the lipid peroxidation concentration. Pretreatment with rutin and hesperidin may cure the liver from doxorubicin-induced injury (Hozayen et al., 2014). Hesperidin also possesses anticancer activity against colon, esophagus, tongue, and urinary bladder carcinoma (Tanaka et al., 2000).

3.5.2.22 Bupleurum chinense

Bupleurum chinense is commonly used herbs for management of exterior syndrome in ancient Chinese medicine. Its leaves have been supposed to be a less medicinally active part in Chinese medicine, but a recent investigation has recommended that the saikosaponins isolated from leaves possess appreciable antioxidant and hepatoprotective activities. An in vitro study revealed that saikosaponins possess radical scavenging efficacy against DPPH radicals and can reduce superoxide anion synthesis in hepatocytes (Liu et al., 2006). It is present in different isoforms and each form may have different biological property. Studies during the past few years on saikosaponin-C have been observed in which the HBV-infected hepatocytes when co-cultured with saikosaponin-C showed significant reduction in HBV antigen expression in growing medium (Chiang et al., 2003). Saikosaponin-B2 also possesses appreciable antiviral activity. In vitro analysis found that it can downregulate early entry of HCV by neutralizing virus particles, preventing its attachment and fusion, and hence inhibiting HCV infection in primary human hepatocytes (Lin et al., 2015).

3.5.2.23 *Cynara scolymus*

Cynara *scolymus* traditionally used for the treatment of digestive disorders, moderate hyperlipidemia, and hepatic bilary diseases. Leaf extract was examined for the presence of luteolin, cynarin, caffeic acid, chlorogenic acid, and other polyphenol and flavonoid compounds, which possess antioxidant activities. *C. scolymus* leaf extract has been useful in hepatic complications (Gebhardt, 1997). It also positively controls the alterations in rat liver enzymes induced by $CC1_4$ along with histopathological impairment (Fallah Huseini et al., 2011). In artichoke extract pretreated rats transaminase activity remarkably declined while histopathological modifications were ameliorated (Mehmetçik et al., 2008). *C. scolymus* also regulated hyperlipidemia, oxidative stress, and abnormal lipid profiles (Heidarian et al., 2011). In rabbit leaf extract neutralized $CC1_4$-induced hepatotoxicity with modification in triglycerides, leukocytes, blood sugar, cholesterol, and a number of erythrocytes (Paunescu et al., 2009). Furthermore, *C. scolymus* appreciably retained the hepatic redox condition as it is exhibited by a noteworthy rise in antioxidant enzyme activities and decline in glutathione accompanied by lipid peroxidation inhibition (LPOI), decreased nitric oxide, protein oxidation, TNF-α, and stabilized membrane permeability in paracetamol-mediated toxicity in rats (Ali et al., 2012).

3.5.2.24 *Polygonum cuspidatum*

Polygonum cuspidatum has been used in folk medicinal system for chronic hepatitis, jaundice, cough, hyperlipidemia, and arthralgia treatment. It is a perennial plant widely grown a in moist environment and ubiquitously distributed worldwide (Zhang et al., 2013a,b). Anthraquinones, resveratrol, and polydatin are some known active compounds present in *P. cuspidatum*. The latest pharmacological evaluation has found that *P. cuspidatum* retain hepatoprotective and antiviral properties (Peng et al., 2013). In vitro analysis has also proved that a *P. cuspidatum* aqueous fraction (30 μg/mL) could decrease the hepatitis-B expression. Whereas, ethanol fraction (10 μg/mL) could suppress the replication of HBV DNA (Chang et al., 2005).

3.5.2.25 *Gynostemma pentaphyllum*

Gynostemma pentaphyllum is a herbaceous climbing plant found in the eastern region of Asia. It is widely used in controlling cholesterol and avoiding obesity and fatty liver in various countries. Wang et al. (2013) observed that its extracts can promote lipid metabolism by downregulating trimethylamine *N*-oxide (TMAO) production and stimulating the excretion of phosphatidylcholine. *G. pentaphyllum* extract efficiently prevents cholesterol and triglycerides accumulation along with neutralizing free radicals in NAFLD model. An additional mechanism associated study suggested that *G. pentaphyllum* stimulates NO production which protects the liver from oxidative damage and modifies the molecular structure of mitochondrial phospholipid cardiolipin (Muller et al., 2012). A clinical trial revealed that intake of *G. pentaphyllum* for 6-months can remarkably decline serum AST, ALP, insulin along with a decrease in body mass index, and insulin resistance index. ALT had reduction of 80.2 to 63.2 and total cholesterol had fallen from 228.5 to 206.1 (Chou et al., 2006). These studies supported that *G. pentaphyllum* can be a suitable medicinal plant for NAFL patients. Recent analysis proposed that ombuine ($C_{17}H_{14}O_7$), a flavonoid extracted from *G. pentaphyllum* significantly suppresses the intracellular triglyceride and cholesterol level in hepatocarcinoma and reduces lipogenic genes expression by stimulating PPARa and ß/d. Moreover, Ombuine also reduces cholesterol level by activating ATP binding cassette cholesterol transporter A1 and G1 protein expression (Malek et al., 2013).

3.5.2.26 *Camellia sinensis*

Camellia sinensis leaves synthesized organic compounds that may protect plants from harmful pathogens, and these compounds are termed as polyphenolic compounds (Friedman, 2007), which include epigallocatechin, caffeine, tannins, catechin, and epicatechin (Wang and Goodman, 1999). *C. sinensis* possesses free-radical scavenging and antioxidant properties (Crespy and Williamson, 2014). The extract of *C. sinensis* and its main polyphenolic compound catechin have medicinal significance against prevention and therapeutics in several diseases (Ostrowska and Skrzydlewska, 2006). It exerts improvement in hepatic metabolic processes by preventing the neutralizing overproduction of ROS and boosting antioxidant defense system capacity. Thus *C. sinensis* extract has protective effects against ethanol toxicity (Lodhi et al., 2014).

3.5.2.27 *Cucurbita pepo*

Cucurbita pepo (pumpkin) are assumed to be a treasure house of antioxidants, polyunsaturated fatty acids (PUFA) and fibers which are known to have hepatoprotective and antiatherogenic properties (Makni et al., 2008). In most countries of the world it is vigorously used in diabetes where it is used internally and superficially for treatment of worms and parasites. Pumpkin is also rich in oleic acid, linoleic acid, and tocopherols and has very high oxidative stability (Stevenson et al., 2007). Linoleic acid, a PUFA present in pumpkin seed oil, is known to increase membrane fluidity and allows for osmosis, intracellular, and extracellular gaseous exchange (Lovejoy, 2002). Pumpkin oil may play an important role in the protection against alcohol-induced hepatotoxicity and oxidative stress. Pretreatment with pumpkin oil may have hepatoprotective effects, which are varied and include oxidation, antilipid peroxidation enhanced detoxification, and protection against glutathione depletion (Abou Seif, 2014).

3.5.2.28 *Zingiber officinale*

Zingiber officinale (Ginger) is extensively used throughout the world in foods as a spice. For centuries, it has been used as a natural medicine for the management of diabetes, catarrh, asthma, rheumatism, stroke, gingivitis, toothache, constipation, and nervous system diseases (Pandey, 2007). In bromobenzene-induced hepatotoxicity, different concentrations of ginger extract cause alteration in antioxidant enzymes, free radicals, biochemical parameters, and drug metabolizing enzymes

(El-Sharaky et al., 2009). The antioxidant molecules prevent ROS production which are potent DNA damaging agent and associated with carcinogenesis (Patel et al., 2000). Ginger root aqueous extract may also causes hepatoprotective activity against aspartame, which causes hepatotoxicity and oxidative stress. Ginger root extract normalized functional hepatic markers (ALT, AST, ALP, γ-GT), serum total protein, albumin and total bilirubin levels, serum LDH activity, α-fetoprotein, and TNF, increased levels of antioxidant enzymes, and reduced levels of MDA (Hozayen and Seif, 2015).

3.5.2.29 Nigella sativa

*N*igella *sativa* contains more than 30 oil compounds. The volatile oil has been showed to possess thymoquinone and number of monoterpenes viz. α-pinene and p-cymene. $CC1_4$ enhanced the LPO and hepatic enzymes (serum or tissue), along with reduction in antioxidant enzyme levels. *N. sativa* administration facilitated decline in LPO and liver enzyme levels and promoted antioxidant enzyme activities (Kanter et al., 2005). The enzymes levels, oxidative stress index, total oxidative status, and myeloperoxidase were remarkably lower in *N. sativa* treated rodents. Whereas, total antioxidant capacity in hepatocytes was appreciably higher as compared to untreated rats (Yildiz et al., 2008). It is effective in management of rheumatism and associated inflammatory problems (Alemi et al., 2013). Furthermore, the antiinflammatory activity of aqueous extract was demonstrated by inhibition efficacy against carrageenan-induced paw edema (Al-Ghamdi, 2001). Thymoquinone pretreated mice significantly suppressed the increased serum enzyme levels along with reduction in MDA content. Thymoquinone expressively increased hepatic nonprotein sulfhydryl (−SH). *N. sativa* helps in prevention of enzymes present in the hepatic neoglucogenesis pathway (Houcher et al., 2007).

3.5.2.30 Naringenin

Naringenin (5,7,4′-thihydroxyflavanone) is a flavanone compound found in citrus fruits and tomatoes. It possesses a vast range of pharmacological properties including hypolipidemic, antihypertensive, antiinflammatory, antioxidant, and antifibrotic functions (Patel et al., 2014). Mira et al. (2002) showed that it has a capacity of reducing the Fe^{3+} and Cu^{2+} but with less potential than quercetin. Chtourou et al. (2015) found that naringenin prevents the reduction of SOD, CAT, GPx, and GSH. Conversely, it also prevented lipid peroxidation, ALT and AST. Relevant results were obtained by Yen et al. (2009) using naringenin alone and in naringenin-loaded nanoparticle system. Both forms demonstrated antioxidant and hepatoprotective activities with caspases 3 and 8 activation. Goldwasser et al. (2011) in their study reported that naringenin activates PPARα and suppresses very low density lipoprotein production without causing lipid accumulation in hepatocytes, in HCV model. Similar results were showed by Cho et al. (2011), who have found that naringenin intake causes significant depletion in the amount of total triglycerides and cholesterol in plasma and the liver of rats. Pretreatment with naringenin-7-*O*-glucoside increased NQO1, ERK, and phosphorylation and translocation of Nrf2 to the nucleus in H9C2 cardiomyocytes and upregulated mRNA expression of GCLC and GCL modifier (Han et al., 2008). Similarly Esmaeili and Alilou (2014), showed that naringenin was capable of attenuating CCl4-induced hepatotoxicity by downregulating TNF-α, iNOS, and cyclooxigenase-2 by increasing Nfr2 and HO-1 expression.

3.6 CONCLUSION

Herbal plants and their formulations based on traditional therapeutic systems, which are used in the management of hepatic abnormalities, are not satisfactory due to their side effects on the kidneys or other cellular organs. Hence it is necessary to develop new hepatoprotective drugs. The current chapter encompasses a comprehensive update and detailed evaluation of extensively used herbal medicine and their formulations against chronic liver diseases (Table 3.1). These plants or bioactive compounds may offer new options to the restricted therapeutic possibilities that occur at present in the management of hepatic abnormalities and should be considered for future studies. These studies also showed that the hepatoprotective activity of herbs is due to the presence of glycosides, flavonoids, triterpenes, and phenolic compounds that are able to prevent oxidative damage, eliminating virus infection, suppressing fibrogenesis, and preventing tumor cell progression. In future studies, basic research along with clinical trials should be designed in such a way that it checks potential toxicity and side effects of herbal medicines. More medicinal herbs and bioactive compounds with no side effects and significant efficacy are expected to be recognized against handling chronic liver abnormalities in the future.

Table 3.1 Mechanism of Action of Plants and Their Bioactive Compounds in Various Hepatic Abnormalities

S. No.	Herbs/ Phytochemicals	Sources	Mode of Actions	References
1.	*Astragalus membranaceus*	*Astragalus membranaceus*	1. Prevent duck HBV DNA replication. 2. Kill serum HBeAg and HBV DNA in chronic viral hepatitis.	Wang et al. (2009)
2.	Baicalin	*Scutellaria baicalensis*	1. Suppress total cholesterol, triglycerides, LDL, ALT and AST level, and promote serum HDL by facilitating of CaMKKβ/AMPK/ACC pathway. 2. Lessening the ischemia/reperfusion damage in alcoholic fatty liver by lessening of myeloid differentiation factor-88 and TLR4 protein expressions and the nuclear translocation of NF-kB after reperfusion.	Xi et al. (2015) Kim et al. (2012)
3.	Berberine	*Coptis chinensis*	1. Falling TG accumulation in the FFA-associated hepatic steatosis. 2. Enhancing insulin resistance of NAFLD by triggering the expression of IRS-2. 3. Decreasing lipogenesis and stimulating lipolysis by averting the expression of SCD1, FAS, SREBP1c, and increasing the expression of CPT1.	Fan et al. (2013) Xing et al. (2011)
4.	*Bupleurum chinense*	*Bupleurum chinense*	1. Possess free radical scavenging potential and inhibiting the superoxide anion formation. 2. Lowering the expression of inflammatory cytokine.	Liu et al. (2006) Lin et al. (2012)

(*Continued*)

Table 3.1 Mechanism of Action of Plants and Their Bioactive Compounds in Various Hepatic Abnormalities *Continued*

S. No.	Herbs/ Phytochemicals	Sources	Mode of Actions	References
5.	Camptothecin	*Camptotheca acuminata*	1. Suppress growth of SMMC-7721 cell by inhibiting cell cycle at the S and G2/M phases, and inducing intrinsic apoptotic pathway. 2. Encourage TRAIL-associated apoptosis in HCC cells by upregulating ROS and ERK/p38 dependent DR5.	Li et al. (2011) Jayasooriya et al. (2014) Zhou et al. (2010)
6,	Curcumin	*Curcuma longa*	1. Preventing cell proliferation and inducing apoptosis on hepatic carcinoma cells.	Wang et al. (2008)
7.	Dioscin	*Dioscorea opposita*	1. Neutralizing oxidative damage, preventing inflammation, cholesterol and triglyceride formations, lowering MAPK phosphorylation levels, promoting β-oxidation of fatty acid, and introducing autophagy to recover fatty liver situations. 2. Protective effect against alcoholic fatty liver by inhibiting alcohol-mediated oxidative damage, inflammatory cytokine production, mitochondrial function, apoptosis, and liver steatosis.	Liu et al. (2015) Xu et al. (2014)
8.	Gallic acid	*Punica granatum*	1. Improving reduced glucose and lipid homeostasis in high-fat diet—associated NAFLD mice. 2. Lowering GSSG content and oxidative stress and promote GSH peroxidase, glutathione, GSH S-transferase, and GSH reductase levels in hepatic tissue.	Chao et al. (2014). Hsu et al. (2007)
9.	Glycyrrhizin	*Glycyrrhiza uralensis*	1. Suppress hepatitis C virus by reducing the activity of phospholipase A2. 2. Downregulating the cytolytic activity of complement system. 3. Trigger the secretion of IL-10 by dendritic cells. 4. Controlling HBV proliferation, decreasing serum ALT.	Matsumoto et al. (2013) Abe et al. (2003) Matsuo et al. (2001)
10.	*Gynostemma pentaphyllum*	*Gynostemma pentaphyllum*	1. Stimulate lipid metabolism by reducing TMAO production and enhancing phosphatidylcholine. 2. Inhibiting cholesterol and triglycerides accumulation along with lowering oxidative stress by promoting NO production and affects the molecular composition of the mitochondrial phospholipid.	Wang et al. (2013) Muller et al. (2012) Chou et al. (2006)
11.	*Panax notoginseng*	*Panax notoginseng*	1. Attenuating the ethanol-mediated lipid accumulation in liver by inhibiting the production of MDA, GSH l and reactive ROS, reducing TNF-alpha and IL-6 levels, as well as enhancing the SOD activity in liver, and abrogated CYP2E1 induction.	Ding et al. (2015) Yang et al. (2009)

Table 3.1 Mechanism of Action of Plants and Their Bioactive Compounds in Various Hepatic Abnormalities *Continued*

S. No.	Herbs/ Phytochemicals	Sources	Mode of Actions	References
			2. Relieve oxidative stress and insulin resistance in NAFLD rats.	
12.	Periplocoside A	*Periploca sepium*	1. Ameliorating of ConeA-induced autoimmune hepatitis by lowering the secretions of (IL)-4, IFN-γ, and ALT.	Wan et al. (2008)
13.	*Phyllanthus niruri*	*Phyllanthus niruri*	1. Inhibiting HBsAg secretion and HBsAg mRNA expression by upregulation of annexin A7. 2. Removing serum HBsAg, HBeAg, and HBV DNA.	Lam et al. (2006) Liu et al. (2001)
14,	*Polygonum cuspidatum*	*Polygonum cuspidatum*	1. Downregulate HBeAg expression and the production of HBV DNA.	Chang et al. (2005)
15.	Puerarin	*Pueraria lobata*	1. Attenuating the CCl_4-induced toxicity in the hepatic cells of hepatic fibrosis rats, mediating antifibrosis effects through modulating the PPAR-gamma expression and inhibiting the PI3K/Akt signal pathway. 2. Mediating antifibrosis effects in hepatic fibrosis rats through downregulating the TNF-alpha and NF-kB expression. 3. Introducing apoptosis in hepatic stellate cells by downregulating bcl-2 mRNA.	Guo et al. (2013) Li et al. (2013) Zhang et al. (2006)
16.	*Salvia miltiorrhiza*	*Salvia miltiorrhiza*	1. Suppressing inflammation in the liver by inhibition of NFkB and p38 signaling. 2. Improve hepatic morphology, decreasing iron deposition as well as inhibiting the expression of type I and type III collagen, TGF-β mRNA, and increase expression of MMP-9 mRNA in the liver.	Yue et al. (2014) Zhang et al. (2013)
17.	*Schisandra chinensis*	*Schisandra chinensis*	1. Anti-HBV activity by suppressing HBV DNA replication.	Xue et al. (2015)
18.	Silymarin	*Silybum marianum*	1. Downregulating the HCV core mRNA and protein expression blocking of HCV entry and transmission by inhibiting microsomal triglyceride transfer protein activity, apolipoprotein B secretion, and infectious virion production into culture supernatants. 2. Decreasing serum transaminases in chronic viral hepatitis but not affecting viral content in vivo. Inhibiting inflammatory by suppressing the proinflammatory cytokine and upregulating the IL-10.	Bonifaz et al. (2009) Wagoner et al. (2010) Mayer et al. (2005). Strickland et al. (2005) Adeyemo et al. (2013)
19.	*Camellia sinensis*	*Camellia sinensis*	1. It promote antioxidant activity and protect liver and various toxic compounds due to the presence of epigallocatechin, caffeine, tannins, catechin, and epicatechin as a bioactive compound.	Wang and Goodman (1999)

(*Continued*)

Table 3.1 Mechanism of Action of Plants and Their Bioactive Compounds in Various Hepatic Abnormalities ***Continued***

S. No.	Herbs/ Phytochemicals	Sources	Mode of Actions	References
20.	*Cucurbita pepo*	*Cucurbita pepo*	1. Linoleic acid, present in it increases membrane fluidity and allow osmosis. 2. Inhibit lipid peroxidation and promote detoxification of liver.	Lovejoy (2002) Abou Seif (2014a)
21.	*Zingiber officinale*	*Zingiber officinale*	1. Root extract promotes serum ALT, AST, ALP, γ-GT level, and increase cellular antioxidant enzymes.	Hozayen and Abou Seif (2014)
22.	Naringenin	Citrus fruits	1. Upregulate the production of SOD, CAT, GPx, and GSH as well as prevent lipid peroxidation. 2. It downregulate TNF-α, iNOS, and cox-2 while promote Nrf2 and HO-1expression in CCl_4-induced oxidative stress.	Chtourou et al. (2015) Motawi et al. (2014)

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REFERENCES

Abe, M., Akbar, F., Hasebe, A., Horiike, N., Onji, M., 2003. Glycyrrhizin enhances interleukin-10 production by liver dendritic cells in mice with hepatitis. J. Gastroenterol. 38 (10), 962–967.

Adeyemo, O., Doi, H., Rajender Reddy, K., Kaplan, D.E., 2013. Impact of oral silymarin on virus- and non-virus-specific t-cell responses in chronic hepatitis c infection. J. Viral. Hepat. 20 (7), 453–462.

Adi, A.H., Alturkmani, H.J., 2013. Physiologically lucky: the role of medical physiology in modern medical education: an interview. Perspect. Med. Educ. 2 (2), 99–103.

Ai, Z.L., Zhang, W.S., Yao, S.K., Xie, B.S., Gao, C., 2011. Effect of baicalin on liver fatty acid binding protein in oxidative stress model in vitro. Chin. J. Hepatol. 19 (12), 927–931.

Akhgar, M.R., Pahlavanzadeh-Iran, S., Lotfi-Anari, P., Faghihi-Zarandi, A., 2011. Composition of essential oils of fruits and leaves of *Prangos ferulacea* (L.) Lindl. growing wild in Iran. Trends Mod. Chem. 1, 1–4.

Alemi, M., Sabouni, F., Sanjarian, F., Haghbeen, K., Ansari, S., 2013. Anti-inflammatory effect of seeds and callus of *Nigella sativa* L. extracts on mix glial cells with regard to their thymoquinone content. AAPS Pharm. Sci. Tech. 14 (1), 160–167.

Al-Ghamdi, M.S., 2001. The anti-inflammatory, analgesic and antipyretic activity of *Nigella sativa*. J. Ethnopharmacol. 76 (1), 45–48.

Ali, Z.Y., Atia, H.A., Ibrahim, N.H., 2012. Possible hepatoprotective potential of *Cynara scolymus*, *Cupressus sempervirens* and *Eugenia jambolana* against paracetamol-induced liver injury: in-vitro and in-vivo evidence. Nat. Sci. 10 (1), 75–86.

Alpers, D.H., Sabesin, S.M., Schiff, L., Schiff, E.R., 1982. Diseases of the Liver. JB Lippincott, Philadelphia, pp. 813−845.
Alsheikh-Ali, A.A., Kuvin, J.T., Karas, R.H., 2004. Risk of adverse events with fibrates. Am. J. Cardiol. 94 (7), 935−938.
Altas, S., Kizil, G., Kizil, M., Ketani, A., Haris, P.I., 2011. Protective effect of DiyarbakIr watermelon juice on carbon tetrachloride-induced toxicity in rats. Food. Chem. Toxicol. 49 (9), 2433−2438.
Amar, P.J., Schiff, E.R., 2007. Acetaminophen safety and hepatotoxicity-where do we go from here? Expert. Opin. Drug. Saf. 6 (4), 341−355.
Amin, Z.A., Abdulla, M.A., Ali, H.M., Alshawsh, M.A., Qadir, S.W., 2012. Assessment of *in vitro* antioxidant, antibacterial and immune activation potentials of aqueous and ethanol extracts of *Phyllanthus niruri*. J. Sci. Food. Agric. 92 (9), 1874−1877.
Arora, N., Goldhaber, S.Z., 2006. Anticoagulants and transaminase elevation. Circulation 113 (15), 698−702.
Bagul, P.K., Middela, H., Matapally, S., Padiya, R., Bastia, T., Madhusudana, K., et al., 2012. Attenuation of insulin resistance, metabolic syndrome and hepatic oxidative stress by resveratrol in fructose-fed rats. Pharmacol. Res. 66 (3), 260−268.
Bhardwaj, S.S., Chalasani, N., 2007. Lipid lowering agents that cause drug induced hepatotoxicity. Clin. Liver. Dis. 11 (3), 597-vii.
Biswas, A., D'Souza, U.J.A., Bhat, S., Damodar, D., 2014. The hepatoprotective effect of *Hibiscus rosa sinensis* flower extract on diet-induced hypercholesterolemia in male albino wistar rats. Int. J. Med. Pharm. Sci. 4 (6), 1−10.
Blaising, J., Levy, P.L., Gondeau, C., Phelip, C., Varbanov, M., Teissier, E., et al., 2013. Silibinin inhibits hepatitis C virus entry into hepatocytes by hindering clathrin-dependent trafficking. Cell Microbiol. 15 (11), 1866−1882.
Boll, M., Weber, L.W., Becker, E., Stampfl, A., 2001. Mechanism of carbon tetrachloride-induced hepatotoxicity. Hepatocellular damage by reactive carbon tetrachloride metabolites. Z. Naturforsch. C 56 (7-8), 649−659.
Bonifaz, V., Shan, Y., Lambrecht, R.W., Donohue, S.E., Moschenross, D., Bonkovsky, H.L., 2009. Effects of silymarin on hepatitis c virus and haem oxygenase-1 gene expression in human hepatoma cells. Liv. Int. Off. J. Int. Assoc. Study Liv 29 (3), 366−373.
Brind, A.M., 2007. Drugs that damage the liver. Medicine. 35 (1), 26−30.
Cai, L., Cao, B., Lin, D., 2015. Effects of traditional Chinese medicine huangqi injection (radix astragali) on random skin flap survival in rats. J. Reconstr. Microsurg. 31 (8), 565−570.
Carlson, M.K., Gleason, P.P., Sen, S., 2001. Elevation of hepatic transaminases after enoxaparin use: case report and review of unfractionated and low molecular weight heparin induced hepatotoxicity. Pharmacotherapy 21 (1), 108−113.
Chang, C.Y., Schaino, T.D., 2007. Review article: drug hepatotoxicity. Aliment. Pharmacol. Ther. 25 (10), 1135−1151.
Chang, J.S., Liu, H.W., Wang, K.C., Chen, M.C., Chiang, L.C., Hua, Y.C., et al., 2005. Ethanol extract of *Polygonum cuspidatum* inhibits hepatitis B virus in a stable HBV-producing cell line. Antiviral. Res. 66 (1), 29−34.
Chao, J., Huo, T.I., Cheng, H.Y., Tsai, J.C., Liao, J.W., Lee, M.S., et al., 2014. Gallic acid ameliorated impaired glucose and lipid homeostasis in high fat diet-induced NAFLD mice. PLoS One 9 (2), e96969.
Checker, R., Patwardhan, R.S., Sharma, D., Menon, J., Thoh, M., Bhilwade, H.N., et al., 2012. Schisandrin B exhibits anti-inflammatory activity through modulation of the redox-sensitive transcription factors Nrf2 and NFkB. Free Radic. Biol. Med. 53 (7), 1421−1430.
Cheng, A.S., Cheng, Y.H., Chiou, C.H., Chang, T.L., 2012. Resveratrol upregulates Nrf2 expression to attenuate methylglyoxal-induced insulin resistance in Hep G2 cells. J. Agric. Food. Chem. 60 (36), 9180−9187.

Chiang, L.C., Ng, L.T., Liu, L.T., Shieh, D.E., Lin, C.C., 2003. Cytotoxicity and anti-hepatitis b virus activities of saikosaponins from bupleurum species. Planta Med. 69 (8), 705−709.

Chitturi, S., George, J., 2002. Hepatotoxicity of commonly used drugs: nonsteroidal anti-inflammatory drugs, antihipertensives, antidiabetic agents, psycotropic drugs. Semin. Liver. Dis. 22 (2), 195−206.

Cho, K.W., Kim, Y.O., Andrade, J.E., Burgess, J.R., Kim, Y.C., 2011. Dietary naringenin increases hepatic peroxisome proliferators activated receptor α protein expression and decreases plasma triglyceride and adiposity in rats. Eur. J. Nutr. 50 (2), 81−88.

Chou, S.C., Chen, K.W., Hwang, J.S., Lu, W.T., Chu, Y.Y., Lin, J.D., et al., 2006. The add-on effects of *Gynostemma pentaphyllum* on nonalcoholic fatty liver disease. Altern. Ther. Health Med. 12 (3), 34−39.

Chtourou, Y., Fetoui, H., Jemai, R., Ben Slima, A., Makni, M., Gdoura, R., 2015. Naringenin reduces cholesterolinduced hepatic inflammation in rats by modulating matrixmetalloproteinases2, 9 via inhibition of nuclear factor κB pathway. Eur. J. Pharmacol. 5 (746), 96−105.

Chvez, E., Reyes-Gordillo, K., Segovia, J., Shibayama, M., Tsutsumi, V., Vergara, P., et al., 2008. Resveratrol prevents fibrosis, NF-kB activation and TGF-β increases induced by chronic CCl4 treatment in rats. J. Appl. Toxicol. 28 (1), 35−43.

Clarke, J.B., Maggs, J.L., Kitteringham, N.R., Park, B.K., 1990. Immunogenicity of amodiaquine in the rat. Int. Arch. Allergy Appl. Immunol. 91 (4), 335−342.

Clarke, J.B., Neftel, K., Kitteringham, N.R., Park, B.K., 1991. Detection of antidrug IgG antibodies in patients with adverse drug reactions to amodiaquine. Int. Allergy Appl. Immunol. 95 (4), 369−375.

Correia, H.S., Batista, M.T., Dinis, T.C., 2007. The activity of an extract and fraction of *Agrimonia eupatoria* L. against reactive species. Biofactors 29 (2-3), 91−104.

Cowan, M.M., 1999. Plant products as antimicrobial agents. Clin. Microb. Rev. 12 (4), 564−582.

Crespy, V., Williamson, G., 2014. A review of the health effects of green tea catechins in vivo animal models. J. Nutr. 134 (12), 3431−3440.

Crowell, J.A., Korytko, P.J., Morrissey, R.L., Booth, T.D., Levine, B.S., 2004. Resveratrol-associated renal toxicity. Toxicol. Sci. 82 (2), 614−619.

Daniel, L., 2009. Global burden of hepatitis C. Liver. Int. 29 (1), 74−81.

Deng, X., Luyendyk, J.P., Ganey, P.E., Roth, R.A., 2009. Inflammatory stress and idiosyncratic hepatotoxicity: hints from animal models. Pharmacol. Rev. 61 (3), 262−282.

Dewick, P.M., 1996. Tumor Inhibition From Plants: Tease and Evans. Elsevier Health Sciences, Philadelphia, PA.

Ding, R.B., Tian, K., Cao, Y.W., Bao, J.L., Wang, M., He, C., et al., 2015. Protective effect of *Panax notoginseng* saponins on acute ethanol-induced liver injury is associated with ameliorating hepatic lipid accumulation and reducing ethanol-mediated oxidative stress. J. Agric. Food. Chem. 63 (9), 2413−2422.

El-Sharaky, A.S., Newairy, A.A., Kamel, M.A., Eweda, S.M., 2009. Protective effect of ginger extract against bromobenzene-induced hepatotoxicity in male rats. Food. Chem. Toxicol. 47 (7), 1584−1590.

Esmaeili, M.A., Alilou, M., 2014. Naringenin attenuates CCl_4 induced hepatic inflammation by the activation of an Nrf2 mediated pathway in rats. Clin. Exp. Pharmacol. Physiol. 41 (6), 416−422.

Faghihzadeh, F., Adibi, P., Rafiei, R., Hekmatdoost, A., 2014. Resveratrol supplementation improves inflammatory biomarkers in patients with nonalcoholic fatty liver disease. Nutr. Res. 34 (10), 837−843.

Fallah Huseini, H., Zareei Mahmoudabady, A., Ziai, S.A., Mehrazma, M., Alavian, S.M., Mehdizadeh, M., et al., 2011. The effects of *Cynara scolymus* L. leaf and *Cichorium intybus* L. root extracts on carbon tetrachloride induced liver toxicity in rats. J. Med. Plants 10 (37), 33−40.

Fan, H., Chen, Y.Y., Bei, W.J., Wang, L.Y., Chen, B.T., Guo, J., 2013. In vitro screening for antihepatic steatosis active components within coptidis rhizoma alkaloids extract using liver cell extraction with hplc analysis and a free fatty acid-induced hepatic steatosis hepg2 cell assay. Evid. Based Complement Altern. Med. 2013, 459390.

Farombi, E.O., Shrotriya, S., Na, H.K., Kim, S.H., Surh, Y.J., 2008. Curcumin attenuates dimethylnitrosamine-induced liver injury in rats through Nrf2-mediated induction of heme oxygenase-1. Food. Chem. Toxicol. 46 (4), 1279−1287.

Formisano, C., Rigano, D., Senatore, F., Bruno, M., Rosselli, S., 2010. Volatile constituents of the aerial parts of white salsify (*Tragopogon porrifolius* L., Asteraceae). Nat. Prod. Res. 24 (7), 663−668.

Fransworth, N.R., 1994. Ethnopharmacology and drug development. In: Chadwick, D.J., Marsh, J. (Eds.), Ethnobotany and the Search for New Drugs, CIBA Foundation Symposium, 185. John Wiley and Sons, Chichester, New York, pp. 42−51.

Friedman, M., 2007. Overview of antibacterial, antitoxin, antiviral and antifungal activities of tea flavonoids and teas. Mol. Nutr. Food. Res. 51 (1), 116−134.

Fujisawa, Y., Sakamoto, M., Matsushita, M., Fujita, T., Nishioka, K., 2000. Glycyrrhizin inhibits the lytic pathway of complement-possible mechanism of its anti-inflammatory effect on liver cells in viral hepatitis. Microbiol. Immunol. 44 (9), 799−804.

Gao, J., Liu, Z.J., Chen, T., Zhao, D., 2014. Pharmaceutical properties of calycosin, the major bioactive isoflavonoid in the dry root extract of radix astragali. Pharm. Biol. 52 (9), 1217−1222.

Gebhardt, R., 1997. Antioxidative and protective properties of extracts from leaves of the artichoke (*Cynara scolymus* L.) against hydroperoxide induced oxidative stress in cultured rat hepatocytes. Toxicol. Appl. Pharmacol. 144 (2), 279−286.

Georgina, E.O., Kingsley, O., Esosa, U.S., Helen, N.K., Frank, A.O., Anthony, O.C., 2011. Comparative evaluation of antioxidant effects of watermelon and orange, and their effects on some serum lipid profile of Wister albino rats. Int. J. Nutr. Metab. 3 (8), 97−102.

Ghany, M., Hoofnagle, J.H., 2005. Approach to the patient with liver disease. In: sixteenth ed. Kasper, D.L., Braunwald, E., Fauci, A.S., Hauser, S.L., Longo, D.L., Jameson, J.L. (Eds.), Harrison's Principles of Internal Medicine, vol. 2. McGraw Hill, New York, pp. 1808−1813.

Giao, M.S., Pereira, C.I., Fonseca, S.C., Pintado, M.E., Malcata, F.X., 2009. Effect of particle size upon the extent of extraction of antioxidant power from the plants *Agrimonia eupatoria*, *Salvia* sp. and *Satureja montana*. Food. Chem. 117 (3), 412−416.

Goldwasser, J., Cohen, P.Y., Lin, W., Kitsberg, D., Balaguer, P., et al., 2011. Naringenin inhibits the assembly and long term production of infectious hepatitis C virus particles through a PPAR mediated mechanism. J. Hepatol. 55 (5), 963−971.

Govind, P., 2011. Medicinal plants against liver diseases. Int. Res. J. Pharm. 2 (5), 115−121.

Green, S.H., 1984. Sodium valproate and routine liver function tests. Arch. Dis. Child. 59 (9), 813−814.

Guo, C., Xu, L., He, Q., Liang, T., Duan, X., Li, R., 2013. Anti-fibrotic effects of puerarin on CCl_4-induced hepatic fibrosis in rats possibly through the regulation of PPAR-γ expression and inhibition of pi3k/akt pathway. Food. Chem. Toxicol. 56, 436−442.

Gupta, R., Gabrielsen, B., Ferguson, F.M., 2005. Nature's medicines: traditional knowledge and intellectual property management. Case studies from the National Institutes of Health (NIH), USA. Curr. Drug Discov. Technol. 2 (4), 203−219.

Gupta, A., Kumar, R., Kumar, S., Pandey, A.K., 2017. Pharmacological aspects of *Terminalia belerica*. In: Mahdi, A.A., Abid, M., Ali-Khan, M.M., Ansari, M.I., Maheshwari, R.K. (Eds.), Molecular Biology and Pharmacognosy of Beneficial Plants. Lenin Media Private Limited, Delhi, India, pp. 52−64.

Guyton, A., Hall, J., 2006. Text Book of Medical Physiology, eleventh ed Saunders, Philadelphia, PA, pp. 859−864.

Han, X., Pan, J., Ren, D., Cheng, Y., Fan, P., Lou, H., 2008. Naringenin 7-O-glucoside protects against doxorubicininduced toxicity in H9c2 cardiomyocytes by induction of endogenous antioxidant enzymes. Food. Chem. Toxicol. 46, 3140−3146.

Haque, A., Tahmina, Afsana, S.K., Sarker, I.R., Hossain, M., Islam, S., 2014. Antioxidant and hepatoprotective effects of aqueous and ethanol extracts of *Dendrophthoe falcate* Linn leaves. Pharmacology online 2014, 90−101.

Hartleb, M., Rymarczyk, G., Januszewski, K., 1999. Acute cholestatic hepatitis associated with provastatin. Am. J. Gastroenterol. 94 (5), 1388−1390.

Heidarian, E., Jafari-Dehkordi, E., Seidkhani-Nahal, A., 2011. Lipid-lowering effect of artichoke on liver phosphatidate phosphohydrolase and plasma lipids in hyperlipidemic rats. J. Med. Plants Res. 5 (19), 4918−4924.

Hosseinimehr, S.J., Nemati, A., 2006. Radioprotective effects of hesperidin against gamma irradiation in mouse bone marrow cells. Brit. J. Radiol. 79 (941), 415−418.

Houcher, Z., Boudiaf, K., Benboubetra, M., Houcher, B., 2007. Effects of methanolic extract and commercial oil of *Nigella sativa* L. on blood glucose and antioxidant capacity in alloxan-induced diabetic rats. Pteridines 18 (1), 8−18.

Hozayen, W.G., Seif, H.S.A., 2015. Chemopreventive effects of *Zinger officinal* extract against aspartame induced nephrotoxicity and oxidative stress in rat model. JIARM 3 (1), 261−273.

Hozayen, W.G., Abou-Seif, H.S., Amin, S., 2014. Protective effects of rutin and/or hesperidin against doxorubicin-induced hepatotoxicity. Int. J. Clin. Nutr. 2 (1), 11−17.

Hsu, C.L., Yen, G.C., 2007. Effect of gallic acid on high fat diet-induced dyslipidaemia, hepatosteatosis and oxidative stress in rats. Br. J. Nutr. 98 (4), 727−735.

Huang, H.L., Wang, Y.J., Zhang, Q.Y., Liu, B., Wang, F.Y., Li, J.J., et al., 2012. Hepatoprotective effects of baicalein against CCL_4-induced acute liver injury in mice. World J. Gastroenterol. 18 (45), 6605−6613.

Iancu, T.C., Shiloh, H., Dembo, L., 1986. Hepatomegaly following short-term high dose steroid therapy. J. Pediatr. Gastroenterol. Nutr. 5 (1), 41−46.

Ibrahim, D., Hong, L.S., Kuppan, N., 2013. Antimicrobial activity of crude methanolic extract from *Phyllanthus niruri*. Nat. Prod. Commun. 8 (4), 493−496.

Jayasooriya, R.G., Choi, Y.H., Hyun, J.W., Kim, G.Y., 2014. Camptothecin sensitizes human hepatoma hep3b cells to trail-mediated apoptosis via ros-dependent death receptor 5 upregulation with the involvement of mapks. Environ. Toxicol. Pharmacol. 38 (3), 959−967.

Kafash Farkhad, N., Farokhi, F., Tukmacki, A., Soltani Band, K., 2012. Hydroalcoholic extract of the root of *Prangos ferulacea* (L.) Lindl can improve serum glucose and lipids in alloxan-induced diabetic rats. Avicenna J. Phytomed. 2 (4), 179−187.

Kafrouni, M.I., Anders, R.A., Verma, S., 2007. Hepatotoxicity associated with dietary supplements containing anabolic steroids. Clin. Gastroenterol. Hepatol. 5 (7), 809−812.

Kanter, M., Coskun, O., Budancamanak, M., 2005. Hepatoprotective effects of *Nigella sativa* L and *Urtica dioica* L on lipid peroxidation, antioxidant enzyme systems and liver enzymes in carbon tetrachloride-treated rats. World J. Gastroenterol. 11 (42), 6684−6688.

Kaplowitz, N., 2004. Drug-induced liver injury. Clin. Infect. Dis. 38 (2), 44−48.

Khalili, H., Dashti-Khavidaki, S., Rasoolinejad, M., Rezaie, L., Etminana, M., 2009. Anti-tuberculosis drugs related hepatotoxicity: incidence, risk factors, pattern of changes in liver enzymes and outcome. Daru 17 (3), 163−167.

Kim, S.J., Moon, Y.J., Lee, S.M., 2010. Protective effects of baicalin against ischemia/reperfusion injury in rat liver. J. Nat. Prod. 73 (12), 2003−2008.

King, P.D., Perry, M.C., 2001. Hepatotoxicity of chemotherapy. Oncologist 6 (2), 162−176.

Konopinski, M., 2009. Influence of intercrop plants and varied tillage on yields and nutritional value of salsify (*Tragopogon porrifolius* L.) roots. Acta Sci. Pol. Hortorum. Cultus 82, 27−36.

Kumar, S., Pandey, A.K., 2013. Chemistry and biological activities of flavonoids: an overview. SciWorldJ 2013, *162750*.

Kumar, S., Gupta, A., Pandey, A.K., 2013. *Calotropis procera* root extract has the capability to combat free radical mediated damage. ISRN Pharmacol. 2013.

Lagouge, M., Argmann, C., Gerhart-Hines, Z., Meziane, H., Lerin, C., Daussin, F., et al., 2006. Resveratrol improves mitochondrial function and protects against metabolic disease by activating SIRT1 and PGC-1alpha. Cell 127 (6), 1109−1122.

Lam, W.Y., Leung, K.T., Law, P.T., Lee, S.M., Chan, H.L., Fung, K.P., et al., 2006. Antiviral effect of *Phyllanthus nanus* ethanolic extract against hepatitis b virus (hbv) by expression microarray analysis. J. Cell. Biochem. 97 (4), 795−812.

Lee, W.M., 2003. Drug-induced hepatotoxicity. N. Engl. J. Med. 349 (5), 474−485.

Li, Q., Qiu, W., Zhu, Q., Zu, Y., Deng, X., Zhao, T., et al., 2011. Camptothecin-20(s)-o-[n-(3'alpha, 12'alpha-dihydroxy-24'-carbonyl-5'beta-cholan)]-lysine, a novel camptothecin analogue, induces apoptosis towards hepatocellular carcinoma smmc-7721 cells. Molecules 16 (9), 7803−7814.

Li, R., Xu, L., Liang, T., Li, Y., Zhang, S., Duan, X., 2013. Puerarin mediates hepatoprotection against ccl4-induced hepatic fibrosis rats through attenuation of inflammation response and amelioration of metabolic function. Food. Chem. Toxicol. 52, 69−75.

Lim, A.Y., Segarra, I., Chakravarthi, S., Akram, S., Judson, J.P., 2010. Histopathology and biochemistry analysis of the interaction between sunitinib and paracetamol in mice. BMC Pharmacol. 10 (14), 1−17.

Lin, L.T., Chung, C.Y., Hsu, W.C., Chang, S.P., Hung, T.C., Shields, J., et al., 2015. Saikosaponin b2 is a naturally occurring terpenoid that efficiently inhibits hepatitis c virus entry. J. Hepatol. 62 (3), 541−548.

Liu, Z.Y., 1991. Progress in pharmacological research on radix astragali. Chin. J Mod. Dev. Tradit. Med. 11, 312−314.

Liu, C.T., Chuang, P.T., Wu, C.Y., Weng, Y.M., Chen, W., Tseng, C.Y., 2006. Antioxidative and in vitro hepatoprotective activity of *Bupleurum kaoi* leaf infusion. Phytother. Res. 20 (11), 1003−1008.

Liu, S., Wei, W., Li, Y., Lin, X., Shi, K., Cao, X., et al., 2014. In vitro and in vivo anti-hepatitis b virus activities of the lignan nirtetralin b isolated from *Phyllanthus niruri* L. J. Ethnopharmacol. 157 (2014), 62−68.

Liu, M., Xu, L., Yin, L., Qi, Y., Xu, Y., Han, X., et al., 2015. Potent effects of dioscin against obesity in mice. Sci. Rep. 5, 7973.

Lodhi, P., Tandan, N., Singh, N., Kumar, D., Kumar, M., 2014. Research article *Camellia sinensis* (L.) Kuntze extract ameliorates chronic ethanol-induced hepatotoxicity in albino rats. Evid. Based. Complement. Alternat. Med. 2014, 7 pages.

Lok, A.S., Heathcote, E.J., Hoofnagle, J.H., 2001. Management of hepatitis B: 2000-summary of a workshop. Gastroenterology 120 (7), 1828−1853.

Lovejoy, J.C., 2002. The influence of dietary fats in insulin resistance. Curr. Diab. Rep. 2 (5), 430−440.

Maffei, F.R., Carini, M., 1980. The inhibitory effect of pyrazinamide on microsomal monooxygenase activities is related to the binding to reduced cytochrome P-450. Pharmacol. Res. Commun. 12 (6), 523−537.

Makni, M., Fetoui, H., Gargouri, N.K., Garoui, M., Jaber, H., Makni, J., et al., 2008. Hypolipidemic and hepatoprotective effects of flax and pumpkin seed mixture rich in u-3 and u-6 fatty acids in hypercholesterolemic rats. Food. Chem. Toxicol. 46 (12), 3714−3720.

Malek, M.A., Hoang, M.H., Jia, Y., Lee, J.H., Jun, H.J., Lee, D.H., et al., 2013. Ombuin-3-o-beta-d-glucopyranoside from *Gynostemma pentaphyllum* is a dual agonistic ligand of peroxisome proliferator-activated receptors α and δ/β. Biochem. Biophys. Res. Commun. 430 (4), 1322−1328.

Mamat, S.S., Kamarolzaman, M.F.F., Yahya, F., Mahmood, N.D., Shahril, M.S., 2013. Methanol extract of *Melastoma malabathricum* leaves exerted antioxidant and liver protective activity in rats. BMC Complement. Altern. Med. 13, 326.

Manov, I., Motanis, H., Frumin, I., Iancu, T.C., 2006. Hepatotoxicity of antiinflammatory and analgesic drugs: ultrastructural aspects. Acta Pharmacol. Sin. 27 (3), 259−272.

Marier, J.F., Vachon, P., Gritsas, A., Zhang, J., Moreau, J.P., Ducharme, M.P., 2002. Metabolism and disposition of resveratrol in rats: extent of absorption, glucuronidation, and enterohepatic recirculation evidenced by a linked-rat model. J. Pharmacol. Exp. Ther. 302 (1), 369–373.

Massumi, M.A., Fazeli, M.R., Alavi, S.H.R., Ajani, Y., 2007. Chemical constituents and antibacterial activity of essential oil of *Prangos ferulacea* (L.) Lindl. fruits. Iran J. Pharm. Sci. 3 (3), 171–176.

Mates, J.M., Segura, J.A., Alonso, F.J., Marquez, J., 2011. Anticancer antioxidant regulatory functions of phytochemicals. Curr. Med. Chem. 18 (15), 2315–2338.

Mayer, K.E., Myers, R.P., Lee, S.S., 2005. Silymarin treatment of viral hepatitis: a systematic review. J. Viral. Hepat. 12 (6), 559–567.

Mehmetçik, G., Ozdemirler, G., Kocak-Toker, N., Cevikbas, U., Uysal, M., 2008. Effect of pretreatment with artichoke extract on carbon tetrachlorideinduced liver injury and oxidative stress. Exp. Toxicol. Pathol. 60 (6), 475–480.

Michael, M.D., Kulkarni, R.N., Postic, C., 2000. Loss of insulin signaling in hepatocytes leads to severe insulin resistance and progressive hepatic dysfunction. Mol. Cell 6 (1), 87–97.

Mira, L., Fernandez, M.T., Santos, M., Rocha, R., Florencio, M.H., Jennings, K.R., 2002. Interactions of flavonoids with iron and copper ions: mechanism for their antioxidant activity. Free. Radic. Res. 36 (11), 1199–1208.

Mishra, A., Sharma, A.K., Kumar, S., Saxena, A.K., Pandey, A.K., 2013. *Bauhinia variegata* leaf extracts exhibit considerable antibacterial, antioxidant and anticancer activities. BioMed Res. Int. 2013, 915436.

Monshouwer, M., Witkamp, R.F., Nijmeijer, S.M., Van-Leengoed, L.A., Verheijden, J.H., Van Miert, A.S., 1995. Infection (*Actinobacillus pleuropneumoniae*)-mediated suppression of oxidative hepatic drug metabolism and cytochrome P450 3A mRNA levels in pigs. Drug Metab. Dispos. 23 (1), 44–47.

Mroueh, M., Daher, C., El-Sibai, M., Tenkerian, C., 2011. Antioxidant and hepatoprotective activity of *Tragopogon porrifolius* methanolic extract. Planta Med. Available from: https://doi.org/10.1055/s-0031-1282460.

Muller, C., Gardemann, A., Keilhoff, G., Peter, D., Wiswedel, I., Schild, L., 2012. Prevention of free fatty acid-induced lipid accumulation, oxidative stress, and cell death in primary hepatocyte cultures by a *Gynostemma pentaphyllum* extract. Phytomed. Int. J. Phytother. Phytopharmacol. 19 (5), 395–401.

Muriel, P., Rivera Espinoza, Y., 2008. Beneficial drugs for liver diseases. J. Appl. Toxicol. 28 (2), 93–103.

Murray, K.F., Hadzic, N., Wirth, S., Bassett, M., Kelly, D., 2008. Drug-related hepatotoxicity and acute liver failure. J. Pediatr. Gastroenterol. Nutr. 47 (4), 395–405.

Newman, D.J., Cragg, G.M., Snader, K.M., 2003. Natural products as sources of new drugs over the period 1981-2002. J. Nat. Prod. 66 (7), 1022–1032.

Nicotera, P., Bellomo, G., Orrenius, S., 1990. The role of Ca^{2+} in cell killing. Chem. Res. Toxicol. 3, 484–494.

Nishimura, Y., Kurata, N., Sakurai, E., Yasuhara, H., 2004. Inhibitory effect of antituberculosis drugs on human cytochrome P450-mediated activities. J. Pharmacol. Sci. 96 (3), 293–300.

Notarbartolo, M., Poma, P., Perri, D., Dusonchet, L., Cervello, M., D'Alessandro, N., 2005. Antitumor effects of curcumin, alone or in combination with cisplatin or doxorubicin, on human hepatic cancer cells. Analysis of their possible relationship to changes in nf-kb activation levels and in IAP gene expression. Cancer Lett. 224 (1), 53–65.

Opara, E.C., Rockway, S.W., 2006. Antioxidants and micronutrients. Dis. Mon. 52 (4), 151–163.

Ostrowska, J., Skrzydlewska, E., 2006. The comparison of effect of catechins and green tea extract on oxidative modifications of LDL in vitro. Adv. Med. Sci. 5, 298–303.

Pal, S., Bhattacharjee, A., Mukherjee, S., Bhattacharya, K., Khowala, S., 2014. Antioxidant and hepatoprotective activity of ethanolic extract of *Alocasia indica* Tuber. Am. J. Phytomed. Clin. Therapeut. 2 (2), 191–208.

Pandey, A.K., 2007. Anti-staphylococcal activity of a pan-tropical aggressive and obnoxious weed *Parthenium histerophorus*: an in vitro study. Natl. Acad. Sci. Lett. 30 (11/12), 383–386.

Parra, J.L., Reddy, K.R., 2003. Hepatotoxicity of hypolipidemic drugs. Clin. Liver. Dis. 7 (2), 415–433.

Patel, K., Krishna, K., Sokoloski, E., Ito, Y., 2000. Preparative separation of curcuminoids from crude curcumin and turmeric powder by pH-zone refining countercurrent chromatography. J. Liq. Chrom. Relat. Tech. 23 (14), 2209–2218.

Patel, K., Singh, G.K., Patel, D.K., 2014. A review on pharmacological and analytical aspects of naringenin. Chin. J. Integr. Med. 24 (7), 551–560.

Paunescu, A., Ponepal, C.M., Draghici, O., Marinescu, A.G., 2009. The CCl4 action upon physiological indices in *Lepus timidus* and the protective role of some substances. An. UO Fasc. Biol 16 (2), 104–107.

Peng, W., Qin, R., Li, X., Zhou, H., 2013. Botany, phytochemistry, pharmacology, and potential application of *Polygonum cuspidatum* sieb.Et zucc.: a review. J. Ethnopharmacol. 148 (3), 729–745.

Perry, M.C., 1992. Chemotherapeutic agents and hepatotoxicity. Semin. Oncol. 19 (5), 551–565.

Pham, T.V., Lu, S., Kaplowitz, N., 1997. Acetaminophen hepatotoxicity. In: Taylor, M.B. (Ed.), Gastrointestinal Emergencies, second ed Williams and Wilkins, Baltimore, pp. 371–388.

Piroth, L., 2005. Liver steatosis in HIV-infected patients. AIDS Rev. 7 (4), 197–209.

Piver, B., Berthou, F., Dreano, Y., Lucas, D., 2001. Inhibition of CYP3A, CYP1A and CYP2E1 activities by resveratrol and other nonvolatile red wine components. Toxicol. Lett. 125 (1-3), 83–91.

Pounds, J.G., Rosen, J.F., 1988. Cellular Ca^{2+} homeostasis and Ca^{2+} mediated cell processes as critical targets for toxicant action: conceptual and methodological pitfalls. Toxicol. Appl. Pharmacol. 94 (3), 331–341.

Reichen, J., 1999. The role of the sinusoidal endothelium in liver function. News Physiol. Sci. 14, 117–121.

Ricaurte, B., Guirguis, A., Taylor, H.C., Zabriskie, D., 2006. Simvastatin-amiodarone interaction resulting in rhabdomyolysis, azotemia, and possible hepatotoxicity. Ann. Pharmacother. 40 (4), 753–757.

Rockstroh, J., Sulkowski, M., Neubacher, D., Mayers, D., Stern, J., 2006. 24-week effcacy of tipranavir boosted with ritonavir in hepatitis B or C coinfected patients. In: 45th Interscience Conference on Antimicrobial Agents and Chemotherapy.

Saleem, T.S.M., Chetty, C.M., Ramkanth, S., Rajan, V.S.T., Kumar, K.M., Gautham, K., 2010. Hepatoprotective herbs-a review. Int. J. Res. Pharm. Sci. 1 (1), 1–5.

Sarich, T.C., Youssef, M., Zhou, T., Adams, S.P., Wall, R.A., Wright, J.M., 1996. Role of hydrazine in the mechanism of isoniazid hepatotoxicity in rabbits. Arch. Toxicol. 70 (12), 835–840.

Saukkonen, J.J., Cohn, D.L., Jasmer, R.M., Schenker, S., Jereb, J.A., Nolan, C.M., et al., 2006. An offcial ATS statement: hepatotoxicity of antituberculosis therapy. Am. J. Respir. Crit. Care. Med. 174 (8), 935–952.

Schimanski, C.C., Burg, J., Mohler, M., Hohler, T., Kanzler, S., Otto, G., et al., 2004. Phenprocoumon-induced liver disease ranges from mild acute hepatitis to (sub-) acute liver failure. J. Hepatol. 41 (1), 67–74.

Seeman, P., 1972. The membrane actions of anesthetics and tranquilizers. Pharmacol. Rev. 24 (4), 583–655.

Seif, H.S.A., 2014. Ameliorative effect of pumpkin oil (*Cucurbita pepo L.*) against alcohol-induced hepatotoxicity and oxidative stress in albino rats. Beni-Suef. Univ. J. Basic Appl. Sci. 3 (3), 178–185.

Senior, J.R., 2008. What is idiosyncratic hepatotoxicity? What is it not? Hepatology 47 (6), 1813–1815.

Sharma, U.K., Sharma, A.K., Gupta, A., Kumar, R., Pandey, A., Pandey, A.K., 2017. Pharmacological activities of cinnamaldehyde and eugenol: antioxidant, cytotoxic and anti-leishmanial studies. Cell Mol. Biol. 63 (6), 73–78.

Sharma, U.K., Kumar, R., Ganguly, R., Gupta, A., Sharma, A.K., Pandey, A.K., 2018. Cinnamaldehyde, an active component of cinnamon provides protection against food colour induced oxidative stress and hepatotoxicity in albino Wistar rats. Vegetos 31 (2), 123–129.

Sharma, U.K., Kumar, R., Gupta, A., Ganguly, R., Singh, A.K., Ojha, A.K., Pandey, A.K., 2019. Ameliorating efficacy of eugenol against metanil yellow induced toxicity in albino Wistar rats. Food Chem. Toxicol. 126, 34–40.

Shear, N., Spielberg, S., 1988. Anticonvulsant hypersensitivity syndrome: in vitro assessment of risk. J. Clin. Invest. 82 (6), 1826−1832.
Sherwood, L., 1997. Human Physiology: From Cells to Systems, third ed Wadsworth, Boston, MA, p. 880.
Shimizu, S., Atsami, R., Itokawa, K., Iwasaki, M., Aoki, T., Ono, C., et al., 2009. Metabolism dependent hepatotoxicity of amodiaquine in glutathione-depleted mice. Arch. Toxicol. 83 (7), 701−707.
Singh, N., Kamath, V., Narasimhamurthy, K., Rajini, P.S., 2008. Protective effects of potato peel extract against carbonatetrachloride-induced liver injury in rats. Environ. Toxicol. Pharmacol. 26 (2), 241−246.
Steele, M.A., Burk, R.F., DesPrez, R.M., 1991. Toxic hepatitis with isoniazid and rifampin. A meta-analysis. Chest 99 (2), 465−471.
Stevenson, D.G., Eller, F.J., Wang, L., Jane, J.L., Wang, T., Inglett, G.E., 2007. Oil and tocopherol content and composition of pumpkin seed oil in 12 cultivars. J. Agric. Food. Chem. 55 (10), 4005−4013.
Stolk, M.F., Becx, M.C., Kuypers, K.C., Seldenrijk, C.A., 2006. Severe hepatic side effects of ezetimibe. Clin. Gastroenterol. Hepatol. 4 (7), 908−911.
Sulkowski, M., Mehta, S., Chaisson, R., Thomas, D., Moore, R., 2004. Hepatotoxicity associated with protease inhibitor-based antiretroviral regimens with or without concurrent ritonavir. AIDS 18 (17), 2277−2284.
Tanaka, T., Kohno, H., Tsukio, Y., Honjo, S., Tanino, M., Miyake, M., 2000. Citrus limonoids obacunone and limonin inhibit azoxymethane-induced colon carcinogenesis in rats. Biofactors 13 (1-4), 213−218.
Tostmann, A., Boeree, M.J., Aarnoutse, R.E., de Lange, W.C.M., Van der ven, A.J., Dekhuijzen, R., 2008. Antituberculosis drug-induced hepatotoxicity: concise up-to-date review. J. Gastroenterol. Hepatol. 23 (2), 192−202.
Van Leth, F., Phanuphak, P., Ruxtrungham, K., Baraldi, E., Miller, S., Gazzard, B., et al., 2004. Comparison of frst-line antiretroviral therapy with regimens including nevirapine, efavirenz, or both drugs, plus stavudine and lamivudine: a randomised open-label trial, the 2NN Study. Lancet 363 (9417), 1253−1263.
Van Rossum, T.G., Vulto, A.G., Hop, W.C., Schalm, S.W., 2001. Glycyrrhizin-induced reduction of alt in European patients with chronic hepatitis C. Am. J. Gastroenterol. 96 (8), 2432−2437.
Wan, J., Zhu, Y.N., Feng, J.Q., Chen, H.J., Zhang, R.J., Ni, J., et al., 2008. Periplocoside A, a pregnane glycoside from *Periploca sepium* bge, prevents concanavalin a-induced mice hepatitis through inhibiting nkt-derived inflammatory cytokine productions. Int. Immunopharmacol. 8 (9), 1248−1256.
Wang, W., Goodman, M.T., 1999. Antioxidant property of dietary phenolic agents in a human LDL-oxidation ex vivo model: interaction of protein binding activity. Nutr. Res. 19 (2), 191−202.
Wang, W.Z., Cheng, J., Luo, J., Zhuang, S.M., 2008. Abrogation of G2/M arrest sensitizes curcumin-resistant hepatoma cells to apoptosis. FEBS Lett. 582 (18), 2689−2695.
Wang, S., Li, J., Huang, H., Gao, W., Zhuang, C., Li, B., et al., 2009. Anti-hepatitis B virus activities of astragaloside IV isolated from radix astragali. Biol. Pharm. Bull. 32 (1), 132−135.
Wang, M., Wang, F., Wang, Y., Ma, X., Zhao, M., Zhao, C., 2013. Metabonomics study of the therapeutic mechanism of *Gynostemma pentaphyllum* and atorvastatin for hyperlipidemia in rats. PLoS One 8 (11), e78731.
Westphal, J.F., Vetter, D., Brogard, J.M., 1994. Hepatic side-effects of antibiotics. J. Antimicrob. Chemother. 33 (3), 387−401.
WHO, 2001. Legal Status of Traditional Medicines and Complementary/Alternative Medicine: A Worldwide Review, 2001. WHO Publishing.
Wieland, T., Faulstich, H., 1978. Amatoxins, phallotoxins, phallolysin and antamanide: the biologically active components of poisonous *Amanita* mushrooms. CRC Crit. Rev. Biochem. 5 (3), 185−260.
Williams, L.D., Burdock, G.A., Edwards, J.A., Beck, M., Bausch, J., 2009. Safety studies conducted on high-purity trans-resveratrol in experimental animals. Food. Chem. Toxicol. 47 (9), 2170−2182.
World Health Organization, 2002. Diet, Physical Activity and Health. World Health Organization, Geneva.
Wu, Y.L., Lian, L.H., Wan, Y., Nan, J.X., 2010. Baicalein inhibits nuclear factor-kB and apoptosis via c-flip and mapk in d-galn/lps induced acute liver failure in murine models. Chem. Biol. Interact. 188 (3), 526−534.

Xing, L.J., Zhang, L., Liu, T., Hua, Y.Q., Zheng, P.Y., Ji, G., 2011. Berberine reducing insulin resistance by up-regulating IRS-2 mRNA expression in nonalcoholic fatty liver disease (NAFLD) rat liver. Eur. J. Pharmacol. 668 (3), 467–471.

Xu, T., Zheng, L., Xu, L., Yin, L., Qi, Y., Xu, Y., et al., 2014. Protective effects of dioscin against alcohol-induced liver injury. Arch. Toxicol. 88 (3), 739–753.

Xue, M., Zhang, L., Yang, M.X., Zhang, W., Li, X.M., Ou, Z.M., et al., 2015a. Berberine-loaded solid lipid nanoparticles are concentrated in the liver and ameliorate hepatosteatosis in db/db mice. Int. J. Nanomed. 10 (1), 5049–5057.

Xue, Y., Li, X., Du, X., Li, X., Wang, W., Yang, J., et al., 2015b. Isolation and anti-hepatitis b virus activity of dibenzocyclooctadiene lignans from the fruits of *Schisandra chinensis*. Phytochemistry 116, 253–261.

Yang, X., Liao, M., Yang, Z., Guo, J., Gao, Q., 2009. Effect of *Panax notoginseng* on genes expression of cyp and gst in liver tissues of rats. China J. Chin. Mater. Med. 34 (18), 2390–2393.

Yen, F.L., Wu, T.H., Lin, L.T., Cham, T.M., Lin, C.C., 2009. Naringeninloaded nanoparticles improve the physicochemical properties and the hepatoprotective effects of naringenin in orallyadministered rats with CCl(4)induced acute liver failure. Pharm. Res. 26 (4), 893–902.

Yildiz, F., Coban, S., Terzi, A., Ates, M., Aksoy, N., Cakir, H., et al., 2008. *Nigella sativa* relieves the deleterious effects of ischemia reperfusion injury on liver. World J. Gastroenterol. 14 (33), 5204–5209.

Yim, S.Y., Lee, Y.J., Lee, Y.K., Jung, S.E., Kim, J.H., Kim, H.J., et al., 2009. Gomisin n isolated from *Schisandra chinensis* significantly induces anti-proliferative and pro-apoptotic effects in hepatic carcinoma. Mol. Med. Rep. 2 (5), 725–732.

Yoon, S.J., Koh, E.J., Kim, C.S., Jee, O.P., Kwak, J.H., Jeong, W.J., et al., 2012. *Agrimonia eupatoria* protects against chronic ethanol-induced liver injury in rats. Food. Chem. Toxicol. 50 (7), 2335–2341.

Yu, C., Shin, Y.G., Chow, A., Li, Y., Kosmeder, J.W., Lee, Y.S., et al., 2002. Human, rat, and mouse metabolism of resveratrol. Pharm. Res. 19 (12), 1907–1914.

Yu, K.Z., Liu, J., Guo, B.L., Zhao, Z.Z., Hong, H., Chen, H.B., et al., 2014. Microscopic research on a multi-source traditional chinese medicine, astragali radix. J. Nat. Med. 68 (2), 340–350.

Yue, S., Hu, B., Wang, Z., Yue, Z., Wang, F., Zhao, Y., et al., 2014. *Salvia miltiorrhiza* compounds protect the liver from acute injury by regulation of p38 and nfkappab signaling in kupffer cells. Pharm. Biol. 52 (10), 1278–1285.

Zeeni, N., Daher, C.F., Saab, L., Mroueh, M., 2014. *Tragopogon porrifolius* improves serum lipid profile and increases short-term satiety in rats. Appetite 72, 1–7.

Zhang, S., Ji, G., Liu, J., 2006. Reversal of chemical-induced liver fibrosis in wistar rats by puerarin. J. Nutr. Biochem. 17 (7), 485–491.

Zhang, J., Ni, J., Chen, Z.H., Li, X., Zhang, R.J., Tang, W., et al., 2009. Periplocoside A prevents experimental autoimmune encephalomyelitis by suppressing IL-17 production and inhibits differentiation of th17 cells. Acta Pharmacol. Sin. 30 (8), 1144–1152.

Zhang, H., Li, C., Kwok, S.T., Zhang, Q.W., Chan, S.W., 2013a. A review of the pharmacological effects of the dried root of *Polygonum cuspidatum* (hu zhang) and its constituents. Evid. Based Complement. Altern. Med. 2013, 13 pages.

Zhang, Y., Xie, Y., Gao, Y., Ma, J., Yuan, J., et al., 2013b. Multitargeted inhibition of hepatic fibrosis in chronic iron-overloaded mice by *Salvia miltiorrhiza*. J. Ethnopharmacol. 148 (2), 671–681.

Zhao, Y., Li, H., Gao, Z., Xu, H., 2005. Effects of dietary baicalin supplementation on iron overload-induced mouse liver oxidative injury. Eur. J. Pharmacol. 509 (2-3), 195–200.

Zhou, L., Li, X., Chen, X., Li, Z., Liu, X., Zhou, S., et al., 2010. *In vivo* antitumor and antimetastatic activities of camptothecin encapsulated with n-trimethyl chitosan in a preclinical mouse model of liver cancer. Cancer Lett. 297 (1), 56–64.

Zhu, Q., Sun, Y., Zhu, J., Fang, T., Zhang, W., Li, J.X., 2014. Antinociceptive effects of sinomenine in a rat model of neuropathic pain. Sci. Rep. 1 (4), 7270.

FURTHER READING

Cottart, C.H., Nivet-Antoine, V., Laguillier-Morizot, C., Beaudeux, J.L., 2010. Resveratrol bioavailability and toxicity in humans. Mol. Nutr. Food. Res. 54 (1), 7–16.

Emim, J.A., Oliveira, A.B., Lapa, A.J., 1994. Pharmacological evaluation of the anti-inflammatory activity of a citrus bioflavonoid, hesperidin, and the isoflavonoids, duartin and claussequinone, in rats and mice. J. Pharm. Pharmacol. 46 (2), 118–122.

Zhang, L., Wang, G., Hou, W., Li, P., Dulin, A., Bonkovsky, H.L., 2010. Contemporary clinical research of traditional chinese medicines for chronic hepatitis B in china: an analytical review. Hepatology 51 (2), 690–698.